

Noisy Parity on Finite and Infinite Cubes

Exact coloring, inspection budgets,
asymptotic laws, and measurable choice

Prepared for Vladimir Reshetnikov

AI-assisted mathematical development and exposition

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Abstract. A configuration consists of finitely or countably many independent fair bits. One bit is deliberately flipped, with a prescribed distribution over its location, and independent noise is then applied to every coordinate. We ask for a single binary labeling that changes its value as often as possible. For homogeneous noise we solve the optimization exactly, classify the optimizing degrees, and determine the exact success attainable with any average inspection budget. A general spectral simulation explains why adaptive inspection cannot improve that budget frontier: a distribution over finite parity functions reproduces every XOR-noise response of a measurable Boolean rule, at no larger expected inspection cost. The infinite form exhibits two different boundaries. Without noise, measurable success has supremum one but no maximizer, while an infinite parity function gives pointwise perfection under a suitable choice principle. Homogeneous positive noise produces finite measurable optimizers. Summable coordinate-dependent noise has a different exact classification: optimization over all coordinate subsets compactifies optimization over finite subsets, and measurable attainment can fail even though a nonmeasurable rule has a measurable, optimal success event. We also derive finite-size transitions, logarithmically periodic corrections for geometric challenge weights, and power-law and lattice corrections for Zipf weights. All principal results for the stated models are proved. The parity and Fourier foundations are classical and explicitly credited. The contribution is a designed problem, its complete solution in the stated models, and derived extensions; historical priority is not established. The report is unrefereed and is not a Lean or Rocq formalization.

[write] **Status and trust boundary, added 5 October 2026, batch 114.** This manuscript is filed in ProveIt as the single-source research report `noisy-parity-cubes` of the research-report collection (provenance, the repository claims checked against the current tree, the two lemmas that the report `measurable-box-games` already proves, reading conventions and the collected non-claims are in Section 1.4). It is AI-assisted (the title page says “AI-assisted mathematical development and exposition”; the PDF author field reads “OpenAI ChatGPT; prepared for Vladimir Reshetnikov”), unrefereed and *not formalized*: no Lean or Rocq declaration states any theorem of this report. *What rests on what.* Every theorem has a written proof; the Fourier and influence machinery is classical [1], and the parity-function and measurable-colouring facts of Section 8 are classical and credited [3, 4, 5]. The exhaustive finite checks (dimensions 1–4) and the 65-digit asymptotic diagnostics of Section 11 support the finite formulas and the signs and sizes of the asymptotic coefficients; no infinite statement rests on a computation. The write adds two remarks with proofs: Remark 5.2 (under strictly decreasing weights no deterministic tree attains the inspection frontier at a non-integer budget below K) and Remark 6.4 (the stability gap of Theorem 6.3 tends to zero as $\eta \downarrow 0$ on a countable cube); it also adds the second-route notes at Lemma 3.2 and after Corollary 3.3, and Section 13.1, which states every open item as a question. Nothing in the source was found to be wrong.

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1 The problem and its mathematical setting

1.1 A puzzle with a finite and an infinite life

Imagine a panel of lamps. A rule assigns every possible lamp configuration one of two labels. The same rule is used whenever the panel is inspected. A referee chooses a random fair configuration, deliberately toggles one lamp, and then applies random faults. The goal is for the two configurations to receive opposite labels.

For a finite panel without faults, the parity of the number of lit lamps solves the problem perfectly. Adding faults changes the optimization: using more lamps makes it more likely that the deliberate toggle is noticed, but also makes the parity vulnerable to more accidental toggles. An infinite panel raises a second issue. Parity of infinitely many lit lamps is not an ordinary convergent sum. There are choice-based parity labelings, but they have strong failures of measurability.

This report asks three precise questions.

1. What is the largest probability that the label changes, for a given distribution of the deliberate toggle and a given noise rate?
2. What can be achieved when the expected number of inspected lamps is bounded, even if the choice of the next lamp depends on earlier observations?
3. Which aspects survive on a countably infinite panel, and where do measurability and choice change attainment of the optimum?

The finite game is a weighted maximum-cut problem on a binary cube. Its infinite form connects this elementary puzzle with Fourier analysis, decision trees, asymptotic optimization, and descriptive set theory.

1.2 Why this is a plausible Glazer–ProveIt intersection

Glazer’s public author biography explicitly identifies choice paradoxes, formal systems, proof-assistant foundations, and puzzles in finite and infinite settings as interests [7]. His box-game paper supplies a direct mathematical connection between prediction and choice [6]. These are reasons to expect interest in the present problem, rather than an assertion of his endorsement or involvement.

Two inspected parts of ProveIt provide concrete points of contact. The module `ThueMorseBooleanCube.lean` proves finite binary-cube product identities, including the signed Thue–Morse product and a binary-weight enumerator [9]. Under binary enumeration, the parity character used here is exactly the finite Thue–Morse sign. The repository’s research report *Countable Information, Uncountably Many Boxes* studies measurable prediction, finite decision trees, and inspection budgets [10]. Its README describes the report as unrefereed and unformalized. We use this as context, not as an assumed theorem dependency.

[write] Checked against the current tree, 5 October 2026. Both statements are correct at the pin and still at the current tree: the two cited files are byte-identical at the pin and at the write. The cited Lean module proves the three finite product identities named in Section 12, and the sign it uses is $(-1)^{\text{binary weight of } n}$, so the parity character of the first m coordinates is that sign under binary enumeration of $\{0,1\}^m$. A neighbouring module, not cited by the source, proves Walsh-character orthogonality on the same dyadic block. The report `measurable-box-games` has five Parts (at the pin as now), and its Part IV proves the two classical lemmas that this report re-proves in Section 3. Details, with declaration names, are in Section 1.4.

The principal results below concern a single labeling evaluated at two correlated configurations. They do not re-prove the repository’s infinite-hat surplus or box-guessing budget theorems. All proofs needed for our game are given here.

1.3 What is classical and what is established here

Walsh expansion, spectral samples, noise operators, and influence identities are standard [1]. Infinite parity functions and their choice-sensitive behavior are also established literature [3, 4]; the measurable chromatic obstruction is classical [5].

The report formulates a weighted noisy-parity puzzle and proves the exact optimization, inspection frontier, optimizer structure, noise asymptotics, and summable-noise attainment results stated below. These are mathematical claims in the specified models, not claims that a named published open problem has been settled. A targeted search found the relevant classical machinery but did not establish historical priority for the precise formulation and assembled results.

1.4 Provenance, repository context and reading conventions

[write] **Added 5 October 2026, batch 114.** *Provenance.* This report is built from one manuscript, the archive `Noisy_Parity_Finite_and_Infinite_Cubes.zip` (474,728 bytes, SHA-256 beginning `c58216c1`; wrapper directory `noisy_parity/`, 13 files, 639,515 bytes unpacked), which arrived in Provelt’s docs/incoming drop zone in commit `2399df2bd` (“New research reports”, 5 October 2026). It is not part of a numbered report bundle. The intake placed it as batch 114 in commit `99053b5d1`. The text is the delivered `noisy_parity.tex` (2,298 lines, 97 labels, all already prefixed `np:`), shipped as `article.tex`; the three figure PDFs, the three programs and their three recorded outputs are shipped byte-identical (`build.sh` in code/). The delivered 32-page `noisy_parity.pdf` is not shipped and is retrievable from the arrival commit; the package has no checksum manifest. The package is “Prepared for Vladimir Reshetnikov” (authorship, kept here, not an author line) and pins Provelt commit `b71545625fedc631edc2ead542abbfba72eb6063` (title page, Appendix B, bibliography), an ancestor of the current tree. Single source, so no merge choice was made.

The repository claims, checked at the current tree. (i) `Analysis/FabiusFunction/Lean/FabiusFunction/ThueMorseBooleanCube.lean` (97 lines, namespace `Fabius`; blob `3527bee8` at the pin and now) proves four theorems: `prod_one_add_eq_sum_powerset` ($\prod_{j \in S} (1 + x_j) = \sum_{T \subseteq S} \prod_{j \in T} x_j$), `prod_one_add_mul_pow` ($\prod_{j < m} (1 + uz^{2^j}) = \sum_{n < 2^m} u^{w(n)} z^n$ over every commutative semiring, w the binary weight), `prod_one_sub_pow_eq_sum_thueMorseSign` ($\prod_{j < m} (1 - z^{2^j}) = \sum_{n < 2^m} \varepsilon(n) z^n$ over every commutative ring) and `sum_pow_binaryWeight_eq_one_add_pow` ($\sum_{n < 2^m} u^{w(n)} = (1 + u)^m$); the sign is $\varepsilon(n) = (-1)^{w(n)}$, the definition of `thueMorseSign` in `FabiusFunction/Arithmetic.lean`. The source’s description is correct, and so is its reading that $\chi_{[m]}$ is ε under binary enumeration of $\{0, 1\}^m$. *Observation of the write, not used by the source:* over $\mathbb{Q}$ or $\mathbb{R}$, the last identity with $u = -\eta/(1 - \eta)$, multiplied by $(1 - \eta)^k$, is $\sum_{n < 2^k} \eta^{w(n)} (1 - \eta)^{k - w(n)} (-1)^{w(n)} = (1 - 2\eta)^k$, that is $\mathbb{E} \chi_S(B) = a^{|S|}$ for $|S| = k$, the noise factor of (5). (ii) The neighbouring module `ThueMorseWalsh.lean` (353 lines, blob `63a333d6` at the pin and now; not cited by the source) proves `sum_neg_one_pow_binaryWeight_land`: for $a < 2^m$, $\sum_{n < 2^m} (-1)^{w(a \& n)}$ is 2^m if $a = 0$ and 0 otherwise (the vanishing of a nontrivial character sum, in binary encoding; the orthogonality $\mathbb{E} \chi_S \chi_T = \mathbf{1}_{\{S=T\}}$ follows from it by $\chi_S \chi_T = \chi_{S \Delta T}$, which that file does not state), the masked character sum `sum_pow_binaryWeight_mul_pow_binaryWeight_land` and the one-point Walsh spectrum `sum_thueMorseSign_mul_walsh` of the full parity. (iii) No Lean or Rocq file of the repository proves Walsh inversion or Parseval for arbitrary functions on a binary cube, or treats influences, decision trees or noise channels (searches for *walsh*, *parseval*, *plancherel*, *influence*, *decision tree*, *noise* in `*.lean` and `*.v`; the Walsh-basis lemma of `Combinatorics/Ramsey/Lean/GowersSzemerédi` concerns Gowers’ Lemma 15.1 over $\mathbb{Z}/N$ and is unrelated). So the statement of Section 12 that its modules are proposed, not present, is still true; (i) and (ii) supply ingredients of its first layer only. This report shares no declaration with any formal file and none of its statements is formalized. (iv) The

report SetTheory/Cardinals/docs/reports/ordinals-and-order-types/measurable-box-games (README blob 04ce7e69 at the pin and now) does study measurable prediction, finite decision trees and inspection budgets (its Parts I, IV and V), and its README calls it “AI-assisted, unrefereed, not formalized”.

Second routes. Two standard lemmas proved here are already proved in Part IV of measurable-box-games [11], where they are credited as classical: Lemma 3.2 (influence at most the expected number of inspections, with its per-coordinate form used in Proposition 3.4) is the second display of its Lemma 64.1, mbg:insp:var-covariance (source 04 of that Part); the statement after Corollary 3.3 that a depth- d tree has Walsh degree at most d is its Lemma 49.1, mbg:qry:lem:degree (source 06). The proofs here are independent one-paragraph arguments of the same kind and are kept as second routes; notes at both places point there. That report works with ± 1 hats and $\chi_A = \prod_{j \in A} x_j$, this one with bits and $\chi_S = (-1)^{\sum_{i \in S} x_i}$; its $\text{Inf}_j(g) = \mathbb{E}(\partial_j g)^2$ equals the flip influence Inf_j used here for ± 1 -valued g . It uses the Walsh expansion and a noise operator for degree and hypercontractive estimates; the write found there no counterpart of the spectral simulation (Theorem 3.1, Corollary 3.3) or of the other results of this report.

See also. ordinals-and-order-types/robust-neutral-choice (batch 114, placed in the same commit, independent source) meets this report only in side sections: choice-built labels invariant under finite changes on $2^{\mathbb{N}}$, their failure of measurability and the Baire property, and a biased-product threshold there ($\sum_i (p_i - \frac{1}{2})^2 = \infty$ for its neutral E_0 -invariant rules) beside Proposition 8.3 here ($\sum_i \min(q_i, 1 - q_i) < \infty$ for total parity labels). These concern different symmetries; neither is an instance of the other. Neither source cites the other.

What the write changed. It added the notes marked [write], this subsection, Remarks 5.2 and 6.4 (each the last numbered statement of its section), Section 13.1 with questions Q13–Q15, labels on the source’s Q1–Q12 (with a ref key on their list, which does not change the output), two preamble definitions and two bibliography entries. No statement, proof or number of the source was changed.

The external sources, as the write read them (5 October 2026). Geschke, Lubarsky and Rahn [3] (manuscript dated March 13, 2015): Lemma 5 proves that a strategy with at most one mistake exists if and only if a parity function exists, Corollary 9(b) that a parity function exists if and only if there is a transversal of E_{ev} -classes within E_0 -classes (the equivalence of Proposition 8.1), and Question 12 asks whether ZF plus a parity function gives a free ultrafilter or an E_0 -transversal. Serafin [4] (arXiv v2, 2 December 2023): Theorem 1, “Assuming the consistency of the existence of an inaccessible cardinal”, gives a model with a parity function and neither a nonprincipal ultrafilter on $\mathbb{N}$ nor an E_0 -transversal (“otherwise known as a Vitali set”); his Section 2 makes “model of set theory” mean a model of ZF+DC; he presents the theorem as the answer to that question. So the two literature statements of Section 8 are accurate as cited. Not read by the write: Conley–Miller [5], O’Donnell [1], OSSS [2], DLMF 25.11 [8] and the Epoch AI biography [7].

[write] Reading conventions, 5 October 2026. The source reuses several letters; each meaning is fixed by its section. No symbol was renamed. “Success” always means *disagreement*, $f(X) \neq f(Y)$; Win is not the probability of a correct label.

Symbol	Meanings (where)	Do not read as
$a, a_i, a(S)$	$a = 1 - 2\eta$ (homogeneous noise); $a_i = 1 - 2\eta_i$ and $a(S) = \prod_{i \in S} a_i$ (summable noise, Section 8)	
A, B	$A = a(\mathbb{N})$ (Theorem 8.6); Fourier coefficients A, B (proofs of Theorems 6.1, 6.3); an independent set (proof of Proposition 8.2); the event $A_n = \{\text{supp } D \subseteq [n]\}$ (Theorem 10.1)	
b_k, b	$b_k = 2P_k - 1$; the modal sequence b (proof of Proposition 8.3); $b_0(\rho)$ (Zipf proof)	
B	the noise vector (B_i) ; the event B_n (proof of Theorem 8.5); the Bernoulli polynomial B_2 (Theorem 9.8)	
C	Zipf constant; $C(f)$, $C_n(f)$ minimum expected tree cost; a cylinder; an E_0 -class with representative r_C	

Symbol	Meanings (where)	Do not read as
D, d	a random finite-support flip vector (Section 10); $D = \alpha/s + 2\epsilon$ (Zipf); the coefficient D of Q8; $d_k = 2ap_{k+1} - (1-a)b_k$; the rounding offset d in $k = Rs + d$	
ϵ, ϵ	the score deficit of Theorem 6.3; the Zipf scale $\epsilon = \eta R$; the zero-noise deficit $\epsilon(f)$	three unrelated quantities
α	the tail index; the spectral mixture $\alpha(S)$; the probabilities α_n of B_n or A_n	
$F_k, F(S)$	$F_k = a^k b_k$; $F(S) = (2p(S) - 1)a(S)$; with equal noise rates $F_k = F([k])$	$F_k > 0$ (false at $k = 0$)
G	the frontier $G(c)$; $G_0(\rho)$ (Zipf proof)	
h	$h_k = p_k/b_k$; a total parity function h (Corollary 10.2); sections $h_{j,z}$	
K, k_η	$K = K_\eta$ the <i>smallest</i> maximizing degree; k_η any chosen maximizing degree (Section 9)	the unique maximizer (an adjacent tie can occur)
$L, \mathcal{L}$	$L(\eta) = 1 - V(\eta)$; label classes $\mathcal{L}_\mu \subseteq \mathcal{L}_{\text{evt}}$; smooth losses $\mathcal{L}_0, \mathcal{L}(x)$	a measurable success event as a measurable label
$M, \mathcal{M}$	$\mathcal{M}$ the optimal family of finite sets (Section 6); $M = \max_S F(S)$; a truncation size M (Section 3); M_0 (Zipf)	
q	biased coordinate probabilities q_i (Proposition 8.3); $q(S)$ (Theorem 10.1)	
R	the shared seed; $R = (C/\eta)^{1/\alpha}$; $R_n = \sum_{i>n} \eta_i$	
s	$s(c)$ (Theorem 9.2); the saddle root $s_\alpha(\epsilon)$	
T, t_k	the operator T_ρ ; the Zipf tail $T(x)$; $t_k = 1 - P_k$	
u	$u_\rho(S)$; $u_\eta = \{x_\eta\}$; the lattice phase $\{Rs\}$; the argument of g_α ; a finite-support translation	
V	optimal scores of different models: $V(p, \eta)$, $V_\rho(c)$, V_n , V_∞ , $V(\eta)$, $V_n(\eta)$, and V in Theorem 8.5	one quantity
η	the noise rate (here); in robust-neutral-choice η is an unrelated tolerance	
Inf_i	flip influence, a probability	a variance of a non-Boolean function

[write] **What is not claimed, collected from the source.** The Fourier, spectral-sample and influence machinery is classical and credited to O'Donnell; infinite parity functions and their choice behaviour are classical (Geschke–Lubarsky–Rahn, Serafin), and the measurable chromatic obstruction is classical (Conley–Miller). The puzzle is designed; historical priority is not established; it is not a named open problem, not refereed, not a Lean or Rocq formalization, and claims no endorsement or involvement of Elliot Glazer. The spectral simulation neither computes f nor preserves correlations of three or more inputs, and computing the spectral distribution is a separate problem. The expected-budget optimum is not identified with the optimum over deterministic trees under an average cost constraint. The restricted parity-choice principle is not equated with AC or with choice for arbitrary pairs; ultrafilter or E_0 -transversal constructions are sufficient, not necessary; no new choice equivalence is inferred. The stability gap may shrink as $\eta \rightarrow 0$ (the write proves it does, Remark 6.4). There is no countably infinite uniform model. The Zipf formula is not read as resolving terms below its remainder. The Decimal checks are not interval certificates, and the plots illustrate proved formulas without being evidence for them. The formalization table proposes modules. The requested LinkedIn page gave no usable content; Glazer's interests are taken from his Epoch AI biography. No repository modification or communication with a named mathematician was part of the deliverable.

2 Rules, notation, and the main answers

Let $I = [n] = \{1, \dots, n\}$, or $I = \mathbb{N} = \{1, 2, \dots\}$, and let

$$\mathcal{X} = \{0, 1\}^I, \quad \mu = \bigotimes_{i \in I} \text{Bernoulli}(1/2).$$

We allow the completion of this probability measure. Addition $\oplus$ means coordinatewise addition modulo two. Write e_i for the configuration supported at the single coordinate i .

Fix positive probabilities p_i with $\sum_i p_i = 1$, ordered so that $p_1 \geq p_2 \geq \dots$. In the infinite case such an ordering exists: there are only finitely many probabilities above any positive threshold. There is no appeal to a uniform probability distribution on $\mathbb{N}$. Define

$$P_k = \sum_{i=1}^k p_i, \quad P_0 = 0, \quad t_k = 1 - P_k, \quad b_k = 2P_k - 1.$$

Definition 2.1 (The homogeneous noisy-parity experiment). Draw $X \sim \mu$, an index J with $\mathbb{P}(J = i) = p_i$, and independent bits $B_i \sim \text{Bernoulli}(\eta)$, all independently. Set

$$Y = X \oplus e_J \oplus B.$$

For a measurable labeling $f : \mathcal{X} \rightarrow \{-1, 1\}$, its score is

$$\text{Win}_{p,\eta}(f) = \mathbb{P}(f(X) \neq f(Y)).$$

The same labeling is used at both inputs. A random labeling uses one shared seed R , independent of X, J, B , and applies f_R twice.

Both X and Y have law μ . When $I = \mathbb{N}$ and $\eta > 0$, infinitely many noise bits are one almost surely. The definition is still ordinary product probability: each coordinate of Y is a measurable function of the sampled variables.

The same-rule condition matters. Allowing two unrelated labelings would permit opposite constants and give a score of one without any inspection. Sharing the seed likewise specifies the resource model: separate fresh output coins would be a different game.

Put $a = 1 - 2\eta$. For $0 \leq \eta < 1/2$, define

$$F_k = a^k b_k, \quad w_k = \frac{1 + F_k}{2}, \quad F_0 = -1, \quad w_0 = 0. \quad (1)$$

The parity on the k heaviest coordinates is

$$\chi_{[k]}(x) = (-1)^{x_1 + \dots + x_k}.$$

Theorem 2.2 (Exact value and exact inspection frontier). For $0 < \eta < 1/2$, let K be the smallest index maximizing w_k , with $0 \leq k \leq n$ in the finite case. Then:

1. The optimal measurable score is $V(p, \eta) = w_K$, and $\chi_{[K]}$ attains it.
2. The maximizing degrees are either K alone, or $K, K + 1$. The index K is finite even when $I = \mathbb{N}$.
3. Let $G(c)$ be linear between successive points (k, w_k) , $0 \leq k \leq K$, and constant with value w_K for $c \geq K$. Among all exact adaptive shared-randomized labeling procedures with expected inspection cost at most c per evaluation, the optimal score is $G(c)$.

Here an exact procedure reads individual coordinates, may adapt its next query to the answers already seen, and terminates with its chosen label almost surely. Cost counts distinct coordinate inspections; repeated queries can be cached. Preprocessing and random-seed generation are free. For noninteger $c < K$, the optimum is attained by choosing the parities of sizes $\lfloor c \rfloor$ and $\lceil c \rceil$ with the corresponding linear interpolation probabilities.

The proof occupies Sections 3–5. The infinity and noise boundaries are as follows.

Setting	Measurable labelings	Additional conclusion
Finite, $\eta = 0$	Perfect parity exists	Inspect every challenged coordinate.
Countably infinite, $\eta = 0$	Supremum 1, not attained	Choice-based parity is pointwise perfect.
Countably infinite, $0 < \eta < 1/2$	A finite optimum is attained	Every measurable optimizer has finite dependence, up to a null set.
Summable coordinate noise	Compact optimization over all subsets	The supremum can require an infinite, nonmeasurable labeling to be attained.

3 A general spectral simulation

3.1 The countable Fourier basis

For a finite subset $S \subseteq I$, define

$$\chi_S(x) = (-1)^{\sum_{i \in S} x_i}.$$

The empty character is identically one. Independence gives $\mathbb{E}_\mu[\chi_S \chi_T] = \mathbf{1}_{\{S=T\}}$. In finite dimension these characters form a basis by counting dimensions.

In countable dimension, functions of the first m bits are spanned by χ_S , $S \subseteq [m]$. Conditional expectations onto these coordinate sigma-algebras converge in L^2 , so their union is dense. Consequently every real $f \in L^2(\mu)$ has the expansion

$$f = \sum_{S \in \text{Fin}(I)} \hat{f}(S) \chi_S, \quad \hat{f}(S) = \mathbb{E}_\mu[f \chi_S], \quad \sum_S \hat{f}(S)^2 = \|f\|_2^2.$$

For a Boolean f , the final sum is one. The family of finite subsets is countable, so its squared coefficients are an ordinary discrete probability distribution.

Theorem 3.1 (Every XOR-noise response is a parity mixture). *Let Z have any probability law ρ on $\mathcal{X}$, independently of $X \sim \mu$, and put $Y = X \oplus Z$. For every measurable Boolean f ,*

$$\text{Win}_\rho(f) = \sum_{S \in \text{Fin}(I)} \hat{f}(S)^2 \frac{1 - \mathbb{E}_\rho \chi_S(Z)}{2}. \quad (2)$$

Thus a shared random parity χ_S , with $\mathbb{P}(S = T) = \hat{f}(T)^2$, reproduces the score of f for every law ρ simultaneously.

Proof. The convolution operator $T_\rho g(x) = \mathbb{E}_\rho[g(x \oplus Z)]$ is an L^2 contraction. Indeed Jensen's inequality followed by invariance of μ gives

$$\|T_\rho g\|_2^2 \leq \mathbb{E}_{\mu, \rho} |g(X \oplus Z)|^2 = \|g\|_2^2.$$

Every character is an eigenfunction:

$$T_\rho \chi_S = (\mathbb{E}_\rho \chi_S(Z)) \chi_S.$$

Apply this to finite Fourier sums and pass to L^2 limits to get

$$\mathbb{E}[f(X)f(Y)] = \langle f, T_\rho f \rangle = \sum_S \hat{f}(S)^2 \mathbb{E}_\rho \chi_S(Z).$$

The series is absolutely convergent because $|\mathbb{E}_\rho \chi_S(Z)| \leq 1$. Finally, $\mathbf{1}_{\{u \neq v\}} = (1 - uv)/2$ for $u, v \in \{-1, 1\}$, proving (2). The distribution of S depends only on f ; therefore the same simulation works for every noise law. $\square$

The theorem matches a two-input response profile. It does not assert that the simulated parity computes the original labeling, nor that it preserves arbitrary three-input or higher correlations. Computing the spectral distribution efficiently is a separate algorithmic question. The squared-coefficient distribution is the classical spectral sample [1]; the operational statement here is its application to this labeling game.

3.2 Why the simulation never uses more expected inspections

Define the flip influence of coordinate i by

$$\text{Inf}_i(f) = \mathbb{P}_\mu(f(X) \neq f(X \oplus e_i)), \quad \mathcal{I}(f) = \sum_i \text{Inf}_i(f).$$

Fourier expansion of $f(x) - f(x \oplus e_i)$ gives

$$\text{Inf}_i(f) = \sum_{S \ni i} \hat{f}(S)^2.$$

Tonelli's theorem therefore gives the possibly infinite identity

$$\mathcal{I}(f) = \sum_S |S| \hat{f}(S)^2. \quad (3)$$

Lemma 3.2 (Pivotality is paid for by inspection). *If an exact adaptive procedure computes f with query count $Q(X)$, then $\mathcal{I}(f) \leq \mathbb{E}Q(X)$.*

Proof. If a computation on x never inspects coordinate i , the same path and output occur on $x \oplus e_i$. Thus, except on the null sets where one of the computations is not exact and terminating,

$$\{f(x) \neq f(x \oplus e_i)\} \subseteq \{\text{the computation on } x \text{ inspects } i\}.$$

Finite flips preserve μ , so these exceptions remain null. Integrating and summing over i proves the inequality. $\square$

[write] Second route, 5 October 2026. This lemma, and its per-coordinate form ($\text{Inf}_i(f)$ is at most the probability that coordinate i is inspected) used in the proof of Proposition 3.4, are the second display of Lemma 64.1 (mbg:insp:var-covariance) in Part IV of the report measurable-box-games [11], proved there by the same path argument for almost surely terminating deterministic inspection algorithms and credited as classical. The proof above is an independent second route; nothing here depends on that report.

Corollary 3.3 (No advantage from adaptive inspection). *For every exact adaptive shared-randomized labeling protocol there is a randomized nonadaptive parity protocol with no larger expected inspection cost and the same score for every independent XOR perturbation law.*

Proof. For a fixed label f , the spectral simulation reads $|S|$ coordinates, whose expectation is $\mathcal{I}(f)$ by (3). Apply Lemma 3.2. For a random label f_R , average the spectral distributions over the shared seed:

$$\alpha(S) = \mathbb{E}_R \hat{f}_R(S)^2.$$

Joint measurability and Tonelli give $\sum_S \alpha(S) = 1$; the same calculation proves both the score and cost assertions. $\square$

This uses an elementary influence bound, not the stronger O’Donnell–Saks–Schramm–Servedio inequality [2]. A uniform maximum depth is also preserved: a depth- d decision tree has Fourier degree at most d , by induction on the root query. Its spectral sample therefore has size at most d .

[write] **Second route, 5 October 2026.** The degree bound is Lemma 49.1 (mbg:qry:lem:degree) in Part IV of measurable-box-games [11], proved there by the same induction on the first query (with ± 1 coordinates); the one-line argument here is a second route.

Proposition 3.4 (When the cost inequality is equality). *In finite dimension let $C(f)$ be the minimum expected number of queries of an exact decision tree under uniform input. Then*

$$C(f) = \mathcal{I}(f) \iff f = \pm \chi_S \text{ for some } S.$$

Proof. A parity has influence and optimal cost both equal to $|S|$. Conversely choose a minimum-cost tree. If f is nonconstant, let i be its root coordinate. Equality in the sum of the nonnegative revelation-minus-influence gaps forces $\text{Inf}_i(f) = 1$, since the root is always read. On a finite cube this means $f(x \oplus e_i) = -f(x)$ for every x , so $f = \chi_{\{i\}}g$ with g independent of i . The two restricted functions are g and $-g$, with equal minimum cost. Optimality of the subtrees gives $C(f) = 1 + C(g)$, while $\mathcal{I}(f) = 1 + \mathcal{I}(g)$. Induction proves that g , and therefore f , is a signed parity. Constant labels are the empty case. $\square$

3.3 The resource optimization for an arbitrary perturbation

Set $u_\rho(S) = (1 - \mathbb{E}_\rho \chi_S(Z))/2$. Corollary 3.3 gives the exact variational formula

$$V_\rho(c) = \sup \left\{ \sum_S \alpha(S) u_\rho(S) : \alpha(S) \geq 0, \sum_S \alpha(S) = 1, \sum_S |S| \alpha(S) \leq c \right\}. \quad (4)$$

In finite dimension this is a finite linear program. An extreme optimal distribution has at most two nonzero masses: if the cost constraint is active, three positive masses have a nontrivial perturbation preserving both total mass and mean cost; if it is inactive, a single mass suffices.

The countable formula is a supremum. Finite mixtures approximate it: replace all sets of size greater than M by the empty set, losing score at most $c/(M+1)$; then approximate the remaining countable distribution by finite support. No universal attainment assertion is needed. For homogeneous noise, the next two sections give an explicit attained solution.

4 Solving the homogeneous puzzle

For $p(S) = \sum_{i \in S} p_i$, independence gives

$$\mathbb{E} \chi_S(e_J \oplus B) = (1 - 2p(S)) a^{|S|}. \quad (5)$$

The score of a parity is consequently

$$\text{Win}_{p,\eta}(\chi_S) = \frac{1 + a^{|S|}(2p(S) - 1)}{2}.$$

There is also a direct explanation. The deliberate flip lies in S with probability $p(S)$. The number of accidental flips in S is even with probability $(1 + a^{|S|})/2$. The two cases give the same formula.

For fixed cardinality k , the score is maximized by selecting the k heaviest coordinates, because $a^k > 0$ and $p(S) \leq P_k$. Theorem 3.1 now proves the upper bound $\text{Win}(f) \leq \sup_k w_k$, and a maximizing character attains it. In countable dimension $P_k \rightarrow 1$, so some F_k is positive, while $F_k \rightarrow 0$ when $0 < a < 1$. A positive maximum is therefore attained among finitely many degrees.

Proposition 4.1 (Exact cutoff and unimodality). *For $0 < \eta < 1/2$, the sequence F_k strictly increases to its maximum, can have one adjacent tie, and then strictly decreases. In finite dimension set $p_{n+1} = 0$. The first maximizing index is*

$$K = \min\{k : 2ap_{k+1} \leq (1 - a)(2P_k - 1)\}. \quad (6)$$

For an index with $b_k > 0$, the equivalent condition for being a maximizing degree is

$$\frac{p_{k+1}}{b_{k+1}} \leq \eta \leq \frac{p_k}{b_k}, \quad (7)$$

where the left bound is zero at $k = n$.

Proof. Write

$$d_k = 2ap_{k+1} - (1 - a)b_k.$$

Then $F_{k+1} - F_k = a^k d_k$, and

$$d_{k+1} - d_k = 2a(p_{k+2} - p_{k+1}) - 2(1 - a)p_{k+1} < 0$$

where both steps exist. Initially $d_0 > 0$; in the infinite case its limit is $-(1 - a) < 0$, and in finite dimension $d_n = -(1 - a) < 0$. This proves the asserted shape and (6). Alternatively,

$$F_k - F_{k-1} = 2a^{k-1}(p_k - \eta b_k).$$

The ratios p_k/b_k are strictly decreasing once $b_k > 0$, yielding (7). A maximizing F_k is positive, as proved above. $\square$

The optimum can therefore be found by a one-dimensional scan. This is more than a numerical minimization prescription: the first change in the sign of d_k certifies global optimality.

Example 4.2 (A concrete infinite puzzle). Let $p_i = 2^{-i}$ and $\eta = 1/10$. Then $a = 4/5$ and

$$(w_0, w_1, w_2, w_3, w_4) = \left(0, \frac{1}{2}, \frac{33}{50}, \frac{173}{250}, \frac{849}{1250}\right).$$

The optimal rule is the parity of the first three bits, with success $173/250 = 0.692$. Inspecting every bit is neither necessary nor a measurable way to compute infinite parity.

5 The exact inspection frontier

Let K be as in Proposition 4.1. Before the maximum,

$$w_{k+1} - w_k = \frac{a^k d_k}{2} > 0.$$

These increments decrease: both a^k and the positive sequence d_k decrease. Hence the function G from Theorem 2.2 is nondecreasing and concave.

Proof of the inspection assertion in Theorem 2.2. By spectral simulation, it suffices to consider a distribution on finite sets S with $\mathbb{E}|S| \leq c$. Its score is bounded by

$$\mathbb{E}w_{|S|} \leq \mathbb{E}G(|S|) \leq G(\mathbb{E}|S|) \leq G(c).$$

The middle inequality is Jensen's inequality. Conversely, if $m = \lfloor c \rfloor$ and $0 \leq c < K$, use $\chi_{[m]}$ with probability $m + 1 - c$ and $\chi_{[m+1]}$ with probability $c - m$. Its expected cost is c , and its score is exactly

$$G(c) = (m + 1 - c)w_m + (c - m)w_{m+1}.$$

For $c \geq K$, use $\chi_{[K]}$. This proves attainment and completes Theorem 2.2. $\square$

A hard limit of m queries has optimum $w_{\min(m, K)}$, by the degree bound for depth- m trees. For a noninteger expected limit, shared randomization implements the exact interpolation. We do not identify this with the different optimization restricted to deterministic trees having only an average cost constraint.

In Example 4.2, a budget of $5/2$ queries gives

$$G(5/2) = \frac{1}{2} \left(\frac{33}{50} + \frac{173}{250} \right) = \frac{169}{250} = 0.676.$$

One fair shared coin selects parity of two or three bits. No adaptive procedure with the same expected budget has a better score.

Proposition 5.1 (Boundary noise rates). *At $\eta = 0$, the expected-budget optimum is the linear interpolation of (k, P_k) , capped at one in finite dimension. At $\eta = 1/2$, it is $\frac{1}{2} \min(c, 1)$.*

Proof. At zero noise the slopes of the interpolation are p_k , hence nonincreasing; the preceding argument applies without a finite saturation index in countable dimension. At noise $1/2$, X and Y are independent and

$$\text{Win}(f) = \frac{1 - (\mathbb{E}f)^2}{2}.$$

All nonempty characters score $1/2$, and the empty character scores zero. Formula (4) gives the claim, attained by mixing the empty and a singleton parity when $c < 1$. $\square$

Remark 5.2 ([write] Deterministic trees miss the frontier between integers, 5 October 2026). Let $0 < \eta < 1/2$, let the weights be strictly decreasing ($p_1 > p_2 > \dots$), and let $0 < c < K$ be a non-integer. Then every labeling f computed by a deterministic (unrandomized) exact adaptive procedure with expected cost $\mathbb{E}Q \leq c$ has $\text{Win}_{p, \eta}(f) < G(c)$. On a finite cube there are finitely many such procedures, so the optimum over deterministic trees with an average cost constraint is strictly below $G(c)$ for every such c . This sharpens the distinction drawn after the proof of Theorem 2.2; how far below, and whether the supremum reaches $G(c)$ on a countable cube, is question Q13 of Section 13.1.

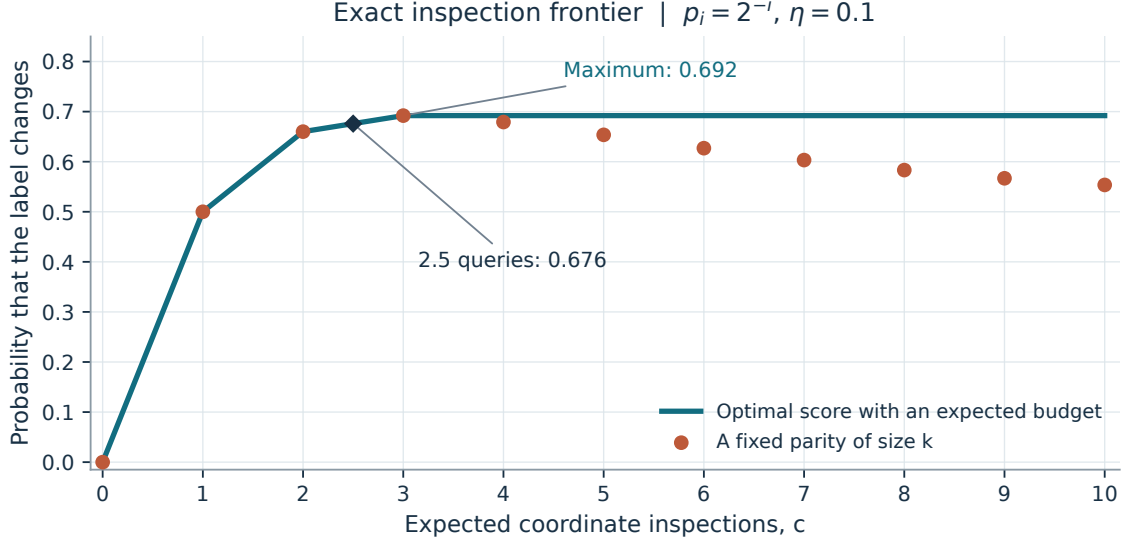


Figure 1: The exact inspection frontier for infinite challenge weights $p_i = 2^{-i}$ at $\eta = 0.1$. Dots show the scores of fixed parity sizes; the solid line is the optimum under an expected query budget. After the third bit, more inspection is unhelpful.

Proof. For finite S put $u(S) = (1 + a^{|S|}(2p(S) - 1))/2$, the score of χ_S (Section 4). Since $a > 0$, $u(S)$ increases strictly with $p(S)$, and $p(S) \leq P_{|S|}$ with equality only for $S = [|S|]$, because the weights are strictly decreasing. Hence $u(S) \leq w_{|S|}$, with equality only for prefix sets. Also $w_k \leq G(k)$ for every k (equality for $k \leq K$; $G(k) = w_K \geq w_k$ for $k > K$). With $\alpha(S) = \hat{f}(S)^2$, Theorem 3.1, Jensen's inequality for the concave G , the identity (3), Lemma 3.2 and the monotonicity of G give

$$\text{Win}(f) = \sum_S \alpha(S) u(S) \leq \sum_S \alpha(S) w_{|S|} \leq \sum_S \alpha(S) G(|S|) \leq G(\mathcal{I}(f)) \leq G(c).$$

Suppose equality held throughout. The first equality forces α to live on prefix sets $[k]$. The increments $w_{k+1} - w_k = a^k d_k / 2$ are positive and strictly decreasing for $k < K$ (both a^k and $d_k > 0$ strictly decrease), and G is constant on $[K, \infty)$; so G is strictly increasing on $[0, K]$ and its slope drops strictly at each integer $1, \dots, K$. Equality in $G(\mathcal{I}(f)) \leq G(c)$ with $\mathcal{I}(f) \leq c < K$ forces $\mathcal{I}(f) = c$. Equality in Jensen's inequality forces the distribution of $|S|$ to lie in one interval on which G is affine, that is in $[m, m+1]$ with $m+1 \leq K$, or in $[K, \infty)$; its mean $\mathcal{I}(f) = c$ is a non-integer below K , so the interval is $[m, m+1]$ with $m = \lfloor c \rfloor$, and both $[m]$ and $[m+1]$ carry positive mass. Thus $f = A\chi_{[m]} + B\chi_{[m+1]} = \chi_{[m]}(A + B\chi_{\{m+1\}})$ with $AB \neq 0$. Evaluating at the two values of $\chi_{\{m+1\}}$, the Boolean condition gives $(A+B)^2 = (A-B)^2 = 1$, hence $AB = 0$, a contradiction. So $\text{Win}(f) < G(c)$. A finite cube has finitely many deterministic exact procedures that never repeat a query (repetitions can be cached, as in Section 2), so the maximum over them is attained and is below $G(c)$. $\square$

6 All optimizers, stability, and ties

Put $\lambda_S = a^{|S|}(1 - 2p(S))$, and let $\lambda_* = -F_K < 0$. Define

$$\mathcal{M} = \{S \in \text{Fin}(I) : \lambda_S = \lambda_*\}.$$

Theorem 6.1 (Optimizer classification). *Assume $0 < \eta < 1/2$. A measurable Boolean labeling is optimal exactly when its Fourier coefficients outside $\mathcal{M}$ vanish. The family $\mathcal{M}$ is finite, so every optimum depends on finitely many coordinates, modulo a null set.*

If all weights are strictly decreasing, the only optima are $\pm\chi_{[K]}$, together with $\pm\chi_{[K+1]}$ when the two degrees tie.

Proof. Equality in the spectral average occurs exactly when all its mass is on least-eigenvalue characters. An optimizing set has an optimizing degree and maximum mass P_k at that degree. There are at most two such degrees. For either one, every coordinate in an optimizing set has weight at least the positive cutoff weight p_k . Only finitely many coordinates meet this condition, so $\mathcal{M}$ is finite.

Strictly decreasing weights give one optimal set at each optimal degree. If there are two degrees, an optimal function has the form

$$f = \chi_{[K]}(A + B\chi_{\{K+1\}}).$$

The Boolean condition implies $(A + B)^2 = (A - B)^2 = 1$, hence $AB = 0$ and $A^2 + B^2 = 1$. The claimed signed parities are the only possibilities. $\square$

Equal coordinate weights permit additional Boolean functions in the optimal eigenspace. This is a genuine exception to a tempting stronger classification.

Example 6.2 (An optimal nonlinear rule). Take

$$p = (3/5, 1/10, 1/10, 1/10, 1/10), \quad \eta = 3/20, \quad a = 7/10.$$

Here $K = 3$. Write $z_i = (-1)^{x_i}$ and set

$$f(x) = \frac{z_1}{2}(z_2z_3 + z_3z_4 + z_4z_5 - z_2z_5).$$

If $z_2 = z_4$, the parenthesized half equals z_2z_3 ; if $z_2 = -z_4$, it equals z_4z_5 . Thus f is Boolean. All four of its Fourier terms use coordinate one and two of the four tied coordinates, so all are optimizing characters. Its score is 6029/10000, although f is not a parity.

Theorem 6.3 (Quantitative stability). *There is a positive gap*

$$\gamma = \inf_{S \notin \mathcal{M}} (\lambda_S - \lambda_*) > 0.$$

If $\text{Win}(f) \geq V - \varepsilon$, then

$$\sum_{S \notin \mathcal{M}} \hat{f}(S)^2 \leq \frac{2\varepsilon}{\gamma}. \quad (8)$$

If $\mathcal{M} = \{S_*\}$, the distance in probability from a suitably signed χ_{S_*} is at most ε/γ . The same distance bound holds to one of the signed optimal parities at a crossover with strictly decreasing weights.

Proof. Large degrees have $|\lambda_S| \leq a^{|S|} \rightarrow 0$, while $\lambda_* < 0$, so they are separated from the minimum. Among the finitely many remaining degrees, the nonoptimal degree maxima have a positive gap. At an optimal degree, a nonmaximizing coordinate set has mass separated from P_k : after the finitely many cutoff ties are accounted for, replacing a selected weight by a strictly lower available weight loses a fixed positive amount. This proves $\gamma > 0$.

The energy excess is

$$2(V - \text{Win}(f)) = \sum_S (\lambda_S - \lambda_*) \hat{f}(S)^2,$$

which proves (8). In the singleton case, write $A = |\hat{f}(S_*)|$. The best sign has mismatch probability $(1 - A)/2 \leq (1 - A^2)/2 \leq \varepsilon/\gamma$.

At a strict-weight crossover put $A = \hat{f}([K])$, $B = \hat{f}([K+1])$. The conditional expectation of $f\chi_{[K]}$ given x_{K+1} is $A + B\chi_{\{K+1\}}$, of absolute value at most one. Thus $|A| + |B| \leq 1$ and $\max(|A|, |B|) \geq A^2 + B^2$. The best signed parity mismatch is at most $(1 - A^2 - B^2)/2$, giving the same bound. $\square$

The gap depends on the weights and the noise rate. In particular, it can shrink as $\eta \downarrow 0$. A fixed-noise stability statement does not by itself give a uniform small-noise conclusion.

Remark 6.4 ([write] The gap tends to zero on a countable cube, 5 October 2026). Let $I = \mathbb{N}$, with all $p_i > 0$, and write $\gamma(\eta)$ and K_η for the gap of Theorem 6.3 and the smallest maximizing degree of Theorem 2.2. Then $0 < \gamma(\eta) < 2p_{K_\eta}$ for every $0 < \eta < 1/2$, and $\gamma(\eta) \rightarrow 0$ as $\eta \downarrow 0$. This proves the source's "can shrink" for every such weight sequence: no positive lower bound for the gap holds uniformly in small η . (On a finite cube the write makes no claim.)

Proof. Write $K = K_\eta$. Since $F_0 = -1$ and the maximum of F_k is positive, $K \geq 1$; and $\lambda_* = -F_K = a^K(1 - 2P_K) = \lambda_{[K]}$. Because $p_j \rightarrow 0$ and $p_K > 0$, some $j > K$ has $p_j < p_K$. Put $S = [K-1] \cup \{j\}$ (so $S = \{j\}$ when $K = 1$). Then $|S| = K$, $p(S) = P_{K-1} + p_j$, and

$$\lambda_S - \lambda_* = a^K[(1 - 2P_{K-1} - 2p_j) - (1 - 2P_K)] = 2a^K(p_K - p_j) \in (0, 2p_K).$$

In particular $\lambda_S \neq \lambda_*$, so $S \notin \mathcal{M}$ and $\gamma(\eta) \leq \lambda_S - \lambda_* < 2p_K$; positivity is Theorem 6.3.

It remains to show $K_\eta \rightarrow \infty$. Fix m . As $\eta \downarrow 0$, $F_k = (1 - 2\eta)^k b_k \rightarrow b_k$ for each of the finitely many $k \leq m+1$, and $b_{m+1} > b_k$ for $k \leq m$ because $b_k = 2P_k - 1$ strictly increases (all $p_i > 0$). Hence for all sufficiently small η , $F_{m+1} > \max_{k \leq m} F_k$, so no $k \leq m$ maximizes and $K_\eta > m$. Since $p_k \rightarrow 0$, $2p_{K_\eta} \rightarrow 0$. $\square$

7 The infinite limit and escape of finite information

Theorem 7.1 (A supremum without a measurable maximizer). *Suppose infinitely many p_i are positive. At zero noise,*

$$\sup_{f \text{ measurable}} \text{Win}_{p,0}(f) = 1, \quad \text{Win}_{p,0}(f) < 1 \text{ for every such } f.$$

Proof. Every finite character has score $p(S) < 1$, and Theorem 3.1 expresses each measurable score as a probability-weighted average of these strictly smaller numbers. Such an average is strictly below one. The prefix parities have scores $P_k \rightarrow 1$, proving the supremum. $\square$

Theorem 7.2 (Every near-perfect sequence loses its finite-coordinate limit). *For a measurable Boolean f , write $\epsilon(f) = 1 - \text{Win}_{p,0}(f)$. Then*

$$\|\mathbb{E}[f \mid X_1, \dots, X_m]\|_2^2 \leq \frac{\epsilon(f)}{1 - P_m}. \quad (9)$$

If $\text{Win}_{p,0}(f_j) \rightarrow 1$, then f_j converges weakly to zero in $L^2(\mu)$. No subsequence converges in measure to a Boolean function.

Proof. The spectral identity gives

$$\epsilon(f) = \sum_S (1 - p(S)) \hat{f}(S)^2.$$

For $S \subseteq [m]$, $1 - p(S) \geq 1 - P_m$. Summing these coefficients proves (9). Each fixed finite-coordinate projection therefore tends to zero, and density of cylinder functions plus $\|f_j\|_2 = 1$ proves weak convergence.

If a subsequence converged in measure to a Boolean g , boundedness would give L^2 convergence: for any $\delta > 0$,

$$\mathbb{E}|f_j - g|^2 \leq \delta^2 + 4\mathbb{P}(|f_j - g| > \delta) \longrightarrow \delta^2.$$

Letting $\delta \downarrow 0$ gives a strong limit g , necessarily also the weak limit zero, contradicting $|g| = 1$. $\square$

This is stronger than observing that one particular sequence of parities fails to converge. Any sequence of increasingly successful measurable labels must lose its mass from each fixed finite-coordinate Fourier space. The obstruction concerns attainment and limits, even though every desired score below one is accessible by a finite rule.

Proposition 7.3 (Quantitative finite approximation). *For $0 \leq \eta < 1/2$, define the n -coordinate challenge distribution $p_i^{(n)} = p_i/P_n$, $i \leq n$, and let V_n be its optimum. Let V_∞ denote the infinite measurable supremum. Then*

$$V_n \downarrow V_\infty, \quad 0 \leq V_n - V_\infty \leq 1 - P_n.$$

Proof. The finite candidate at degree $k \leq n$ is

$$\frac{1 + a^k(2P_k/P_n - 1)}{2}.$$

For fixed k , it decreases as n increases. The new candidate at $k = n + 1$ has advantage a^{n+1} , bounded by the old full-parity advantage a^n . Hence V_n decreases. Every infinite candidate with $k \leq n$ is bounded by the corresponding finite one, and every $k > n$ has advantage at most $a^k \leq a^n$. Thus $V_n \geq V_\infty$. Finally each finite candidate exceeds its infinite counterpart by

$$a^k P_k (P_n^{-1} - 1) \leq 1 - P_n.$$

Take the maximum to get the upper error bound. $\square$

At zero noise all V_n equal one, but the infinite supremum is unattained. At fixed positive noise, Theorem 2.2 supplies an actual finite measurable optimizer. These statements separate numerical convergence of optimal values from convergence or existence of optimizing rules.

8 Choice, measurability, and summable noise

The distinction between a finite cube and the full space $\mathcal{X} = \{0, 1\}^{\mathbb{N}}$ involves both probability and choice. In this section all product measures are completed. We retain $p_i > 0$ and $\sum_i p_i = 1$, write $p(S) = \sum_{i \in S} p_i$, and take labels with values in $\{-1, 1\}$. The fair product measure on $\mathcal{X}$ is denoted by μ .

8.1 The classical parity principle

Two points $x, y \in \mathcal{X}$ are E_0 -equivalent if they differ in finitely many coordinates. They are E_{ev} -equivalent if that difference is finite and has even cardinality. A *total parity label* is a function $f : \mathcal{X} \rightarrow \{-1, 1\}$ satisfying

$$f(x \oplus e_i) = -f(x) \quad (x \in \mathcal{X}, i \in \mathbb{N}). \quad (10)$$

Repeated application gives the corresponding identity for every finite set of flipped coordinates.

Proposition 8.1 (Classical parity construction). *The axiom of choice implies the existence of a total parity label. More precisely, over ZF its existence is equivalent to the following restricted choice statement: from each pair of E_{ev} -classes contained in a single E_0 -class, one can select one member of that pair.*

Proof. Each E_0 -class splits into exactly two E_{ev} -classes: a one-coordinate flip exchanges them, and the parity of a finite difference determines to which one a point belongs. Selecting one of these two classes and assigning it the value 1, with value -1 on the other, gives (10). Conversely, a total parity label is constant on each E_{ev} -class and takes opposite values on the two classes inside any E_0 -class, so its 1-class supplies the selection.

For the usual construction from full choice, select a representative r_C of each E_0 -class C and put

$$f(x) = (-1)^{|\text{supp}(x \oplus r_C)|}, \quad x \in C.$$

The support in this expression is finite. Flipping one coordinate changes its cardinality by one modulo two. $\square$

These constructions and the equivalence with the optimal sequential infinite hat strategy are classical; see Geschke, Lubarsky, and Rahn [3]. The restricted choice statement above must not be replaced by full choice, or by choice for arbitrary families of pairs. Nor should the sufficient constructions using a nonprincipal ultrafilter or an E_0 -transversal be mistaken for necessary conditions. Serafin proved, relative to the consistency of an inaccessible cardinal, that a model of ZF+DC can contain a parity function while containing neither a nonprincipal ultrafilter on $\mathbb{N}$ nor an E_0 -transversal [4]. This answers the corresponding question in the earlier paper negatively.

Proposition 8.2 (Classical regularity obstruction). *In the graph on $\mathcal{X}$ whose edges join points differing in one coordinate, every μ -measurable independent set has measure zero, and every independent set with the Baire property is meager. Consequently the graph has no countable proper measurable coloring and no countable proper Baire-measurable coloring. In particular, no total parity label is measurable or Baire-measurable.*

Proof. Suppose that a measurable independent set A has positive measure. Approximation by finite-coordinate cylinders, or martingale convergence, gives a nonempty cylinder C with $\mu(A \cap C) > \mu(C)/2$. Choose i beyond the coordinates specified by C . The map $x \mapsto x \oplus e_i$ preserves C and its measure. Therefore $A \cap C$ and its image intersect, producing an edge with both endpoints in A , a contradiction.

If an independent set A has the Baire property and is nonmeager, there is a nonempty cylinder C on which it is comeager. A coordinate flip beyond the specified prefix maps C homeomorphically to itself. The two comeager sets $A \cap C$ and its image intersect, giving the same contradiction. Finally, countably many null sets cannot cover $\mathcal{X}$, and countably many meager sets cannot cover the nonempty Baire space $\mathcal{X}$. Apply these statements to the color classes. $\square$

This stronger chromatic formulation is also classical; Conley and Miller explicitly record the example [5]. Together with Theorem 7.1, it displays two different facts: measurable zero-noise scores can approach one arbitrarily closely, whereas no countable regular proper coloring exists.

8.2 Measurable labels and measurable success events

A nonmeasurable label need not make every question about it nonmeasurable. At zero noise, a total parity label succeeds for every (x, j) , so its success event is the whole sample space. For later use, we therefore distinguish the class $\mathcal{L}_\mu$ of μ -measurable labels from the class $\mathcal{L}_{\text{evt}}$ of arbitrary total labels for which

$$\{(x, j, z) : f(x) \neq f(x \oplus e_j \oplus z)\} \quad (11)$$

is measurable for the completed joint experiment. Its probability $\text{Win}(f)$ is defined for $f \in \mathcal{L}_{\text{evt}}$. Since the perturbed configuration also has fair product distribution, $\mathcal{L}_\mu \subseteq \mathcal{L}_{\text{evt}}$. No integral of a nonmeasurable label alone is intended.

Proposition 8.3 (Parity under a biased product measure). *Let f be any total parity label, and let ν_q be the product measure with independent Bernoulli(q_i) coordinates, where $0 < q_i < 1$. Then f is measurable for the completion of ν_q if and only if*

$$\sum_{i=1}^{\infty} \min(q_i, 1 - q_i) < \infty. \quad (12)$$

Proof. Choose the modal sequence b by setting $b_i = 1$ when $q_i > 1/2$ and $b_i = 0$ otherwise. If (12) holds, the first Borel–Cantelli lemma says that ν_q -almost every x differs from b in finitely many coordinates. The class $[b]_{E_0}$ is countable and Borel, and on it

$$f(x) = f(b)(-1)^{|\text{supp}(x \oplus b)|}.$$

This restriction has a Borel extension to $\mathcal{X}$, for example by assigning the value 1 outside the class. The extension agrees with f almost everywhere, proving measurability for the completed measure.

For the converse, translation by b preserves the parity property and replaces q_i by $\min(q_i, 1 - q_i)$. Thus suppose that $0 < q_i \leq 1/2$, $\sum_i q_i = \infty$, and f is measurable. Fix a cylinder C depending on the first n coordinates. Integrating first over the coordinates $n < i \leq m$ and using the parity identity gives

$$\left| \int_C f d\nu_q \right| \leq \nu_q(C) \prod_{i=n+1}^m (1 - 2q_i). \quad (13)$$

Indeed, for fixed coordinates outside this block, the label on the block is its finite parity times a sign independent of the block. All finite block atoms have positive probability, so the necessary sections of a completed-measurable function are measurable almost everywhere, and finite Fubini applies.

Every tail sum of the q_i diverges. The product in (13) therefore tends to zero, using $1 - 2q_i \leq \exp(-2q_i)$, with the conclusion immediate if a factor vanishes. All cylinder integrals are zero. It follows that $\mathbb{E}_{\nu_q}[f \mid x_1, \dots, x_n] = 0$ for every n . Martingale convergence now gives $f = 0$ almost surely, contradicting $|f| = 1$. $\square$

Corollary 8.4 (Homogeneous noise excludes total parity events). *If $0 < \eta < 1$ and the noise coordinates are independent Bernoulli(η) variables, no total parity label belongs to $\mathcal{L}_{\text{evt}}$.*

Proof. Suppose that (11) is measurable. Fix j with $p_j > 0$. Fubini implies that for μ -almost every x its section as a function of z is measurable for the completed noise law. However, for every fixed x, j the sign

$$z \mapsto f(x)f(x \oplus e_j \oplus z)$$

is itself a total parity label on the noise space. Proposition 8.3 rules out its measurability, since $\sum_i \min(\eta_i, 1 - \eta_i) = \infty$. The success indicator determines this sign, so the same obstruction applies to the event section. $\square$

Thus an arbitrary total parity cannot simply be inserted into the positive homogeneous-noise objective. The obstruction concerns the success event itself. The next result shows exactly how summable noise changes this conclusion.

8.3 An exact classification for summable heterogeneous noise

Assume for the remainder of this section that

$$0 \leq \eta_i < \frac{1}{2}, \quad \sum_i \eta_i < \infty, \quad a_i = 1 - 2\eta_i. \quad (14)$$

For every subset $S \subseteq \mathbb{N}$, including infinite subsets, define

$$a(S) = \prod_{i \in S} a_i, \quad F(S) = (2p(S) - 1)a(S), \quad A = a(\mathbb{N}). \quad (15)$$

All these products are positive: only finitely many η_i can be bounded away from zero, their factors are nonzero, and the remaining logarithms have absolutely summable magnitude.

Theorem 8.5 (Summable-noise optimization). *Under (14), let*

$$M = \max_{S \subseteq \mathbb{N}} F(S), \quad V = \frac{1 + M}{2}.$$

Then the maximum defining M exists, and the following hold.

1. $M = \sup\{F(S) : S \subseteq \mathbb{N} \text{ is finite}\}.$
2. $\sup_{f \in \mathcal{L}_\mu} \text{Win}(f) = V.$
3. *A measurable label attains V if and only if at least one maximizing subset S is finite.*
4. *In ZFC, $\max_{f \in \mathcal{L}_{\text{evt}}} \text{Win}(f) = V$; a maximizing label can be constructed from a maximizing subset S .*

Proof. Compactness of the scalar problem. Identify subsets of $\mathbb{N}$ with Cantor space. Write $S_n = S \cap \{1, \dots, n\}$, $P_n = \sum_{i \leq n} p_i$, and $R_n = \sum_{i > n} \eta_i$. The elementary inequality $1 - \prod_j (1 - t_j) \leq \sum_j t_j$ gives

$$0 \leq a(S_n) - a(S) \leq 2R_n.$$

Consequently

$$|F(S) - F(S_n)| \leq 2(1 - P_n) + 2R_n, \quad (16)$$

uniformly in S . Each truncated function is continuous, so F is continuous. Compactness supplies a maximizer; finite subsets are dense, proving the first assertion.

Measurable labels. The Walsh calculation of Theorem 3.1 applies with $a(S)$ in place of the homogeneous factor. More explicitly, for each finite S the character $\chi_S(x) = (-1)^{\sum_{i \in S} x_i}$ has channel eigenvalue $a(S)(1 - 2p(S))$. The finite Walsh characters form an orthonormal basis of $L^2(\mu)$, and hence every measurable label satisfies

$$2 \text{Win}(f) - 1 = \sum_{S \text{ finite}} F(S) \hat{f}(S)^2, \quad \sum_{S \text{ finite}} \hat{f}(S)^2 = 1. \quad (17)$$

This bounds its score by V , and finite characters approximate V by the first assertion. If a finite subset maximizes, its character attains V . Conversely, if every finite S has $F(S) < M$, the right side of (17) is a countable probability-weighted average of numbers strictly smaller than M and is itself strictly smaller than M . This proves the second and third assertions.

Construction for an arbitrary maximizing subset. Choose representatives r_C of the E_0 -classes and, for a fixed $S \subseteq \mathbb{N}$, set

$$f_S(x) = (-1)^{|S \cap \text{supp}(x \oplus r_C)|}, \quad x \in C. \quad (18)$$

For any vector u of finite support this satisfies

$$f_S(x)f_S(x \oplus u) = (-1)^{|S \cap \text{supp}(u)|}. \quad (19)$$

By Borel–Cantelli, Z has finite support almost surely. Therefore, almost surely,

$$f_S(X)f_S(X \oplus e_J \oplus Z) = (-1)^{\mathbf{1}_{\{J \in S\}} + |S \cap \text{supp}(Z)|}.$$

The success event thus agrees almost surely with an event depending only on (J, Z) and is measurable for the completed joint law. Independence, followed by bounded convergence for finite noise truncations, gives correlation $(1 - 2p(S))a(S)$ and score $(1 + F(S))/2$. A maximizing S consequently realizes V in $\mathcal{L}_{\text{evt}}$.

Upper bound for arbitrary labels with measurable events. This step cannot use the Fourier coefficients of f , since they need not exist. Let $f \in \mathcal{L}_{\text{evt}}$ and condition the experiment on

$$B_n = \{J \leq n, Z_i = 0 \text{ for all } i > n\}, \quad \alpha_n = \mathbb{P}(B_n) = P_n \prod_{i>n} (1 - \eta_i).$$

Summability gives $\alpha_n \rightarrow 1$. Conditional on B_n , the challenge weights are p_i/P_n for $i \leq n$, and the first n noise coordinates retain their original independent distributions.

This conditional experiment has only finitely many pairs (j, z) . For each pair of positive conditional probability, the corresponding success section

$$h_{j,z}(x) = \mathbf{1}_{\{f(x) \neq f(x \oplus e_j \oplus z)\}}$$

is μ -measurable. Indeed, summable noise is supported on the countable set of finite-support vectors; every pair used here is a positive atom of the (J, Z) law, so completed joint measurability implies measurability of that section. Pairs of probability zero do not enter the conditional average.

Average the conditional success function over the 2^n translations $x \mapsto x \oplus u$ supported on $\{1, \dots, n\}$. Translation invariance of μ preserves its integral. At every fixed x , however, the resulting orbit average is precisely the finite-cube score of the Boolean function $u \mapsto f(x \oplus u)$. This is a finite function irrespective of the regularity of f . The finite Fourier theorem bounds the orbit average pointwise by

$$W_n = \frac{1}{2} \left[1 + \max_{S \subseteq \{1, \dots, n\}} a(S) \left(\frac{2p(S)}{P_n} - 1 \right) \right].$$

For every candidate in this maximum, its conditional score differs from its original score by

$$a(S)p(S) \left(\frac{1}{P_n} - 1 \right) \leq 1 - P_n.$$

Thus $W_n \leq V + 1 - P_n$, and the unconditional score satisfies

$$\text{Win}(f) \leq \alpha_n W_n + (1 - \alpha_n).$$

Taking $n \rightarrow \infty$ proves $\text{Win}(f) \leq V$. Combined with the construction above, this completes the proof. $\square$

The theorem identifies the same numerical value in the two regularity classes, but attainment can differ. Its compact parameter space consists of arbitrary subsets of coordinates. The Fourier modes available to a measurable label are indexed by finite subsets. This is the source of the attainment criterion, rather than a failure of compactness of the scalar objective.

8.4 Exactly when full infinite parity is optimal

Theorem 8.6 (Full-parity criterion). *Under (14), the subset $S = \mathbb{N}$ maximizes F if and only if*

$$\eta_i \leq p_i \quad \text{for every } i. \quad (20)$$

When this condition holds, the maximizing subsets are exactly

$$\mathbb{N} \quad \text{and} \quad \mathbb{N} \setminus \{j\} \quad \text{for those } j \text{ with } \eta_j = p_j.$$

Consequently the measurable-label supremum is $(1 + A)/2$ and is unattained, whereas a total parity label attains it in $\mathcal{L}_{\text{evt}}$.

Proof. Deleting a single coordinate gives

$$F(\mathbb{N} \setminus \{j\}) - F(\mathbb{N}) = \frac{2A}{a_j}(\eta_j - p_j). \quad (21)$$

Thus optimality of $\mathbb{N}$ requires (20).

Conversely, for any $S \subseteq \mathbb{N}$ and $j \notin S$, direct algebra gives

$$F(S \cup \{j\}) - F(S) = 2a(S) [(p_j - \eta_j) + 2\eta_j p(\mathbb{N} \setminus (S \cup \{j\}))]. \quad (22)$$

Under (20) this is nonnegative. Adjoin the omitted coordinates successively and use continuity of F to obtain $F(S) \leq F(\mathbb{N}) = A$.

If at least two coordinates are omitted, choose any omitted j . When $p_j > \eta_j$, the first term in the brackets of (22) is positive. When $p_j = \eta_j$, their common value is positive, and the remaining omitted coordinate has positive challenge mass, making the second term positive. Such an S therefore cannot maximize. A single omission ties exactly when $\eta_j = p_j$, by (21). This proves the stated classification. All of these maximizing sets are infinite, so the attainment claims follow from Theorem 8.5. $\square$

Remark 8.7 (Positive noise at every coordinate without a measurable optimizer). For any positive probability sequence p , set

$$\eta_i = \frac{p_i^2}{2}, \quad a_i = 1 - p_i^2.$$

Then $0 < \eta_i < p_i$ and $\sum_i \eta_i < \infty$. The unique maximizing subset is $\mathbb{N}$, and the common optimal value is

$$V = \frac{1}{2} \left(1 + \prod_{i=1}^{\infty} (1 - p_i^2) \right).$$

Every finite parity is strictly suboptimal, and (17) excludes every other measurable optimizer as well. For the concrete choice $p_i = 2^{-i}$,

$$\prod_{i=1}^{\infty} (1 - 4^{-i}) \approx 0.6885375371203397, \quad V \approx 0.8442687685601699.$$

A total choice-based parity label attains this value with a measurable success event. The strict inequalities $\eta_i > 0$ at every coordinate therefore do not suffice for the finite optimizer conclusion proved for homogeneous noise.

The probabilistic distinction is concrete. Homogeneous positive noise flips infinitely many coordinates almost surely and takes the experiment outside the finite-difference relation. Summable noise flips only finitely many coordinates almost surely, so the parity identities remain valid on almost every experiment. Proposition 8.3 and Theorem 8.5 make these two statements precise without assigning probabilities to nonmeasurable events.

9 Small-noise laws and finite-size transitions

The exact spectral solution permits a more quantitative question: how many coordinates should a successful rule retain as the noise decreases, and how quickly does its failure probability vanish? The answers depend strongly on the tail of the distinguished-coordinate distribution. Exponential tails produce logarithmic degrees and a periodic second-order correction; power law tails produce fractional powers of the noise. A finite uniform model has a separate saturation transition.

Throughout this section $0 < \eta < 1/2$ and the weights are arranged in nonincreasing order. We retain the notation

$$a = 1 - 2\eta, \quad t_k = 1 - P_k, \quad b_k = 1 - 2t_k, \quad F_k = a^k b_k, \quad w_k = \frac{1 + F_k}{2},$$

and write

$$V(\eta) = \max_k w_k, \quad L(\eta) = 1 - V(\eta).$$

The symbol k_η denotes any specifically chosen maximizing degree; it need not be the smallest maximizing degree $K_\eta = K(p, \eta)$ from Theorem 2.2. An elementary identity controls every degree transition:

$$F_k - F_{k-1} = 2a^{k-1}(p_k - \eta b_k). \quad (23)$$

On the range $b_k > 0$, the ratios $h_k = p_k/b_k$ strictly decrease. Consequently a maximizing degree satisfies

$$\frac{p_{k+1}}{b_{k+1}} \leq \eta \leq \frac{p_k}{b_k}, \quad (24)$$

where the left endpoint is zero at the last coordinate of a finite model. At a crossover two adjacent degrees may be optimal. These statements concern degrees; ties between coordinate weights or within an extremal Fourier eigenspace can allow additional maximizing functions.

9.1 Uniform finite models and the saturation transition

Proposition 9.1 (Uniform weights). *For $p_1 = \dots = p_n = 1/n$, one maximizing degree is*

$$k_\eta = \min \left\{ n, \left\lfloor \frac{n + \eta^{-1}}{2} \right\rfloor \right\}, \quad V_n(\eta) = \frac{1}{2} \left[1 + (1 - 2\eta)^{k_\eta} \frac{2k_\eta - n}{n} \right]. \quad (25)$$

If $(n + \eta^{-1})/2$ is an integer at most n , that integer and its predecessor tie. Otherwise the maximizing cardinality is unique. In particular, all n coordinates are optimal whenever $\eta \leq 1/n$.

Proof. For $k > n/2$ one has $b_k = (2k - n)/n$ and $h_k = 1/(2k - n)$. Inserting this into (24) gives the formula and the tie rule. The last degree becomes optimal at $h_n = 1/n$. $\square$

Theorem 9.2 (A finite-size transition). *Fix $c > 0$ and let $\eta = c/n$. As $n \rightarrow \infty$,*

$$\frac{k_\eta}{n} \longrightarrow s(c) = \begin{cases} 1, & 0 < c \leq 1, \\ (1 + c^{-1})/2, & c \geq 1, \end{cases} \quad (26)$$

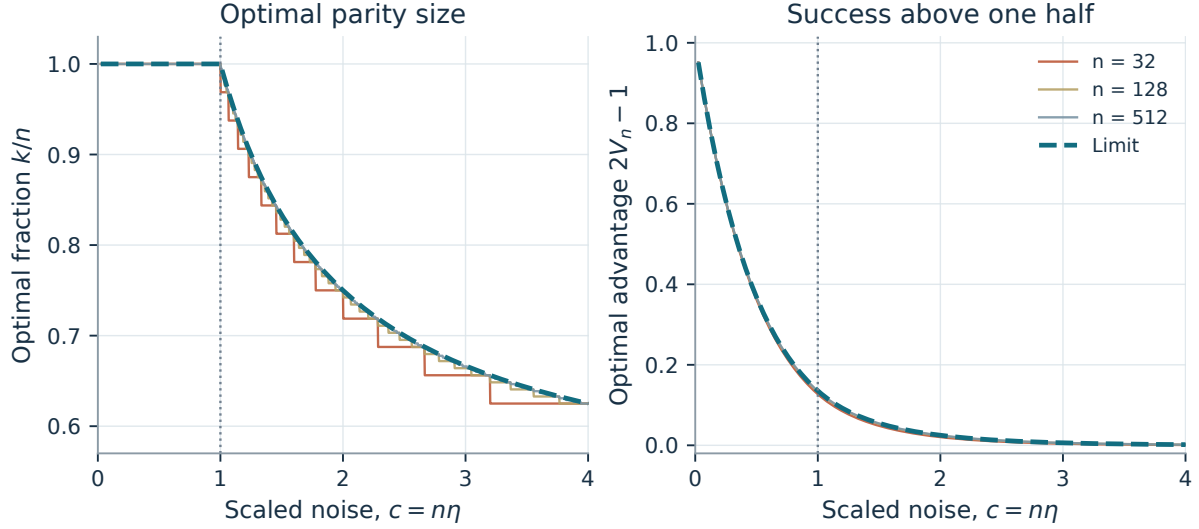


Figure 2: Uniform finite challenge distributions under the scaling $\eta = c/n$. Left: the fraction of coordinates in an optimal parity. Right: the optimal advantage $2V_n - 1$. Curves for finite n approach the limits in Theorem 9.2; $c = 1$ is the saturation threshold.

and

$$2V_n(c/n) - 1 = \mathcal{A}(c) + O_c(n^{-1}), \quad \mathcal{A}(c) = \begin{cases} e^{-2c}, & 0 < c \leq 1, \\ c^{-1}e^{-c-1}, & c \geq 1. \end{cases} \quad (27)$$

The function $\mathcal{A}$ is continuously differentiable at $c = 1$, but its one-sided second derivatives there are $4e^{-2}$ and $5e^{-2}$.

Proof. The exact formula gives $k_\eta = ns(c) + O(1)$, while $\log(1 - 2c/n) = -2c/n + O_c(n^{-2})$. Thus the limiting advantage is $e^{-2cs}(2s - 1)$ evaluated at $s = s(c)$. Equivalently, maximizing this expression on $1/2 \leq s \leq 1$ gives the unconstrained critical point $s = (1 + c^{-1})/2$; it first enters the admissible interval at $c = 1$. Direct differentiation proves the final assertion. $\square$

For fixed $\eta > 0$, in contrast, $V_n(\eta) \rightarrow 1/2$ exponentially. Writing $k = n/2 + u$, the exact advantage, once the upper endpoint is inactive, is

$$2V_n(\eta) - 1 = \frac{2a^{n/2}}{n} \max_{u \in \Lambda_n, u > 0} ua^u, \quad \Lambda_n = \begin{cases} \mathbb{Z}, & n \text{ even}, \\ \mathbb{Z} + \frac{1}{2}, & n \text{ odd}. \end{cases}$$

The prefactor therefore remembers the parity of n . Neither limit constitutes a countably infinite uniform model: there is no uniform probability distribution on a countably infinite coordinate set.

9.2 Geometric tails and a periodic second term

Theorem 9.3 (Geometric weights). Suppose $p_k = (1 - r)r^{k-1}$, where $0 < r < 1$, and put $\ell = -\log r$. Define

$$x_\eta = \frac{1}{\ell} \log \frac{1 - r + 2\eta r}{\eta r}, \quad u_\eta = \{x_\eta\}. \quad (28)$$

Then $k_\eta = \lfloor x_\eta \rfloor$ is optimal; if x_η is an integer, its predecessor also ties. In particular,

$$V(\eta) = \frac{1}{2} \left[1 + (1 - 2\eta)^{\lfloor x_\eta \rfloor} (1 - 2r^{\lfloor x_\eta \rfloor}) \right].$$

As $\eta \downarrow 0$,

$$L(\eta) = \frac{\eta}{\ell} \log \frac{1}{\eta} + \eta \Phi_r(u_\eta) + O_r\left(\eta^2 \log^2 \frac{1}{\eta}\right), \quad (29)$$

where

$$\Phi_r(u) = \frac{\log((1-r)/r)}{\ell} - u + \frac{r}{1-r} e^{\ell u}, \quad 0 \leq u \leq 1. \quad (30)$$

The endpoint identity $\Phi_r(0) = \Phi_r(1)$ makes Φ_r a continuous periodic function. The error estimate is uniform through degree crossovers.

Proof. Here $t_k = r^k$ and

$$h_k = \frac{(1-r)r^{k-1}}{1-2r^k}.$$

Solving $h_k = \eta$ for real k gives (28); monotonicity and (23) give the exact integer rule. Set $c_r = \log((1-r)/r)$. Uniformly in $u_\eta \in [0, 1)$,

$$\begin{aligned} k_\eta &= \frac{\log(1/\eta) + c_r}{\ell} - u_\eta + O_r(\eta), \\ r^{k_\eta} &= \frac{\eta r}{1-r+2\eta r} e^{\ell u_\eta} = \eta \frac{r}{1-r} e^{\ell u_\eta} + O_r(\eta^2). \end{aligned}$$

Since $k_\eta = O_r(\log(1/\eta))$,

$$a^{k_\eta} = 1 - 2\eta k_\eta + O_r\left(\eta^2 \log^2(1/\eta)\right).$$

Substitution into $L = (1 - a^{k_\eta})/2 + a^{k_\eta} r^{k_\eta}$ yields (29). The endpoint identity follows by direct evaluation. $\square$

Thus $k_\eta \sim \ell^{-1} \log(1/\eta)$ and $L(\eta) \sim (\eta/\ell) \log(1/\eta)$. The next coefficient is generally periodic rather than constant. Because the periodic extension of Φ_r is Lipschitz, one may replace u_η by $\{\ell^{-1} \log((1-r)/(\eta r))\}$ in (29) without changing its error term.

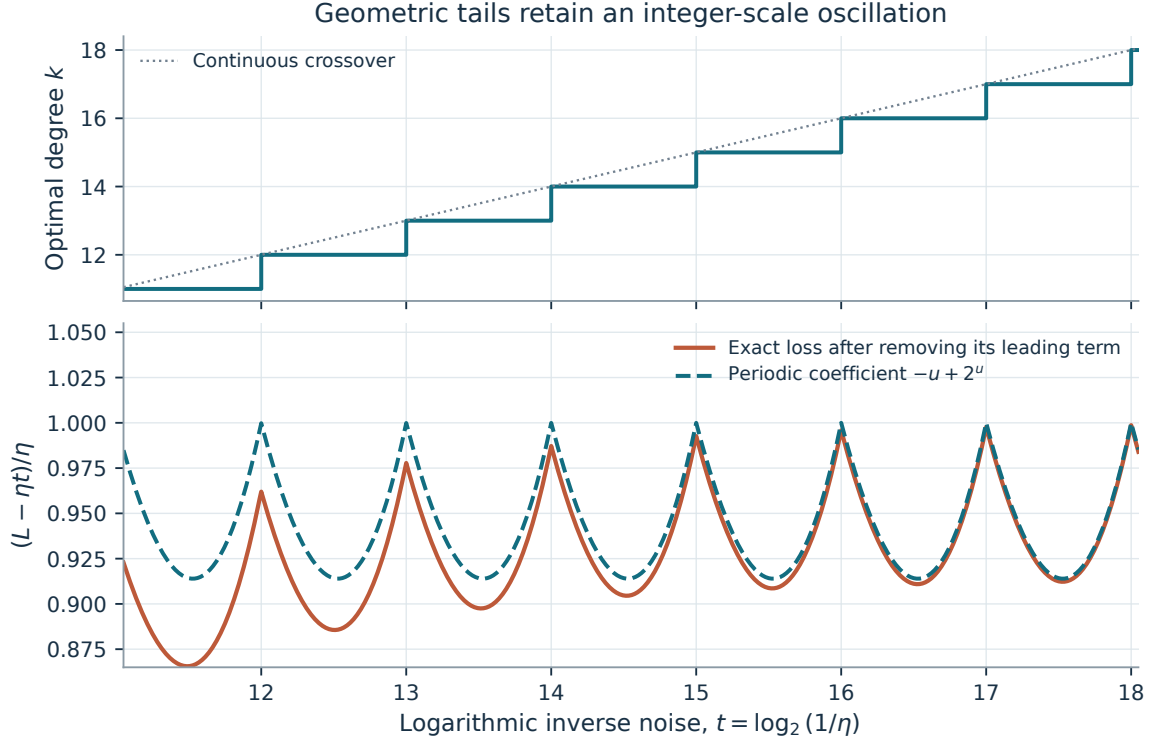


Figure 3: For $p_i = 2^{-i}$, the optimal degree grows in integer steps as η decreases. The lower panel removes the leading term from the optimal failure probability and displays the periodic correction $\Phi_{1/2}(u) = -u + 2^u$. The finite-noise residual approaches this correction as the error term in Theorem 9.3 decreases.

9.3 Regularly varying tails and degree stability

A positive sequence is regularly varying with index $-\alpha$ if

$$\frac{p_{\lfloor tk \rfloor}}{p_k} \longrightarrow t^{-\alpha} \quad \text{for each fixed } t > 0.$$

The following result applies to the standing nonincreasing ordering of the weights.

Theorem 9.4 (The power law for optimal failure). *Suppose (p_k) is regularly varying with index $-\alpha$, where $\alpha > 1$. For any maximizing degree k_η ,*

$$p_{k_\eta} \sim \eta, \quad t_{k_\eta} \sim \frac{k_\eta p_{k_\eta}}{\alpha - 1}, \quad L(\eta) \sim \frac{\alpha}{\alpha - 1} \eta k_\eta. \quad (31)$$

Moreover, for each $c > 0$, $k_{c\eta}/k_\eta \rightarrow c^{-1/\alpha}$. In particular, if $p_k \sim Ck^{-\alpha}$, then, with $R = (C/\eta)^{1/\alpha}$,

$$k_\eta \sim R, \quad L(\eta) \sim \frac{\alpha}{\alpha - 1} C^{1/\alpha} \eta^{(\alpha-1)/\alpha}. \quad (32)$$

Proof. We first justify the tail estimate

$$\frac{t_k}{kp_k} \longrightarrow \int_1^\infty x^{-\alpha} dx = \frac{1}{\alpha - 1}. \quad (33)$$

On every compact interval of positive x , the ratios $p_{\lfloor xk \rfloor}/p_k$ converge uniformly: they are nonincreasing in x and their pointwise limit is continuous. Riemann sums therefore give

the truncated integral. To control the remaining tail, choose $0 < \delta < \alpha - 1$ and a fixed dilation $q > 1$. Regular variation, with harmless adjustments of integer endpoints, bounds the contribution of the j th block $[q^j k, q^{j+1} k]$ by a constant times $q^{j(1-\alpha+\delta)}$. The resulting geometric series is summable, uniformly for all sufficiently large k , and its tail tends to zero. This proves (33).

The crossover rule implies $k_\eta \rightarrow \infty$. Monotonicity and regular variation also imply $p_{k+1}/p_k \rightarrow 1$; hence the two sides of (24) are asymptotically equal to p_k . It follows that $p_{k_\eta} \sim \eta$. Summability and monotonicity imply $k p_k \rightarrow 0$, so $\eta k_\eta \rightarrow 0$ and

$$L(\eta) = \frac{1 - a^{k_\eta}}{2} + a^{k_\eta} t_{k_\eta} = \eta k_\eta + t_{k_\eta} + o(\eta k_\eta + t_{k_\eta}).$$

Together with (33), this proves (31). Finally, regular variation brackets the index at which p_k crosses $c\eta$ between fixed multiples of k_η approaching $c^{-1/\alpha}$. This proves the inverse scaling assertion and its pure-power specialization. $\square$

For example, if $p_k \sim Ck^{-\alpha}(\log k)^\beta$, then direct inversion gives

$$k_\eta \sim C^{1/\alpha} \alpha^{-\beta/\alpha} \eta^{-1/\alpha} (\log(1/\eta))^{\beta/\alpha},$$

and the failure probability is asymptotic to $\alpha \eta k_\eta / (\alpha - 1)$.

Proposition 9.5 (The degree-regret profile). *Assume $p_k \sim Ck^{-\alpha}$, and let $R = (C/\eta)^{1/\alpha}$. For each fixed $u > 0$,*

$$\frac{1 - w_{\lfloor uR \rfloor}}{\eta R} \longrightarrow g_\alpha(u) = u + \frac{u^{1-\alpha}}{\alpha - 1}. \quad (34)$$

The convergence is uniform on compact subintervals of $(0, \infty)$. The unique minimum is $g_\alpha(1) = \alpha / (\alpha - 1)$, and

$$g_\alpha(1 + d) - g_\alpha(1) = \frac{\alpha}{2} d^2 + O_\alpha(d^3).$$

Consequently, if a sequence of parity degrees has failure probability asymptotic to $L(\eta)$, its degree divided by R tends to one.

Proof. For $k = \lfloor uR \rfloor$, expand $(1 - a^k)/2 \sim \eta k$ and use $t_k \sim Ck^{1-\alpha}/(\alpha - 1)$. This proves (34); differentiation gives the minimum and its quadratic expansion. A degree with $k/R \rightarrow 0$ has a tail loss too large on the scale ηR . A degree with $k/R \rightarrow \infty$ has a noise loss too large on that scale, whether or not ηk tends to zero. Every near-optimal sequence is therefore confined to a compact range of positive k/R , where the unique minimum forces $k/R \rightarrow 1$. $\square$

The Fourier formula extends this conclusion beyond parity rules. If a Boolean rule has success deficit $o(\eta R)$, then, for every fixed $\delta > 0$, its Fourier mass on sets satisfying $||S|/R - 1| \geq \delta$ tends to zero. Indeed, the preceding proof supplies a positive gap of order ηR outside that degree range, and a set of size k cannot outperform the k largest weights. Averaging those gaps with the squared Fourier coefficients proves the assertion.

9.4 Exact Zipf weights: an envelope and second-order laws

Second-order statements require more than a leading tail equivalence. We now impose the exact distribution

$$p_k = Ck^{-\alpha}, \quad C = \zeta(\alpha)^{-1}, \quad \alpha > 1,$$

and use the scales

$$R = (C/\eta)^{1/\alpha}, \quad \varepsilon = \eta R. \quad (35)$$

Both ε and R^{-1} tend to zero; keeping them distinct reveals the competition between tail corrections and integer rounding.

Theorem 9.6 (A controlled asymptotic envelope). *Let $s = s_\alpha(\varepsilon) > 1$ solve*

$$s^\alpha - \frac{2\varepsilon}{\alpha - 1}s = 1, \quad (36)$$

and put

$$M_0 = s^{-\alpha} e^{-2\varepsilon s}, \quad E_0 = \frac{1 - M_0}{2}. \quad (37)$$

Then

$$L(\eta) = E_0 + \eta M_0 \left(\varepsilon s - \frac{1}{2} \right) + O_\alpha(\eta/R). \quad (38)$$

In particular,

$$L(\eta) = \frac{\alpha}{\alpha - 1}\varepsilon - \frac{\alpha + 1}{\alpha - 1}\varepsilon^2 - \frac{\eta}{2} + O_\alpha\left(\varepsilon^3 + \eta\varepsilon + \frac{\eta}{R}\right). \quad (39)$$

Proof. The root in (36) is unique for $s \geq 1$. The implicit-function theorem at $(s, \varepsilon) = (1, 0)$ makes it analytic for small ε . Extend the tail to real $x > 0$ by $T(x) = C\zeta(\alpha, x + 1)$. Euler–Maclaurin, or the Hurwitz zeta expansion in [8], gives

$$T(x) = \frac{C}{\alpha - 1}x^{1-\alpha} - \frac{C}{2}x^{-\alpha} + \frac{C\alpha}{12}x^{-\alpha-1} + O_\alpha(x^{-\alpha-3}). \quad (40)$$

The half term is negative because the discrete tail begins at $k + 1$.

Let H be the large real solution of $CH^{-\alpha}/(1 - 2T(H)) = \eta$. Monotonicity makes $\lfloor H \rfloor$ an optimal degree, with its predecessor also permitted at an integer tie. Writing $H = R\rho$, equation (40) gives

$$\rho^\alpha - \frac{2\varepsilon}{\alpha - 1}\rho = 1 - \eta + O_\alpha(\eta/R).$$

Comparison with (36) yields

$$H = Rs - \frac{\varepsilon}{D} + O_\alpha(\eta + \eta\varepsilon), \quad D = \frac{\alpha}{s} + 2\varepsilon. \quad (41)$$

Here the derivative of the left side at s equals D , by the saddle equation. Every maximizing integer thus has $k = Rs + d$ with $|d| \leq 1 + O_\alpha(\varepsilon)$.

For the evaluation, define the smooth leading loss

$$\mathcal{L}_0(x) = \frac{1}{2} \left[1 - e^{-2\eta x} \left(1 - \frac{2C}{\alpha - 1}x^{1-\alpha} \right) \right].$$

Its derivative vanishes at $x_0 = Rs$, and $\mathcal{L}_0(x_0) = E_0$. Put

$$b_0(\rho) = 1 - \frac{2\varepsilon}{\alpha - 1}\rho^{1-\alpha}, \quad G_0(\rho) = e^{-2\varepsilon\rho} \left[\varepsilon\rho b_0(\rho) - \frac{1}{2}\rho^{-\alpha} \right].$$

Uniformly for ρ near one, the exact smooth extension $\mathcal{L}(x) = (1 - a^x(1 - 2T(x)))/2$ satisfies

$$\begin{aligned} \mathcal{L}(R\rho) &= \mathcal{L}_0(R\rho) + \eta G_0(\rho) + \frac{\alpha\eta}{12R}e^{-2\varepsilon\rho}\rho^{-\alpha-1} \\ &\quad + O_\alpha(\eta^2\varepsilon + \eta/R^3). \end{aligned} \quad (42)$$

This follows by substituting (40) and

$$a^{R\rho} = e^{-2\varepsilon\rho} [1 - 2\eta\varepsilon\rho + O_\alpha(\eta^2\varepsilon)].$$

At $\rho = s$, the saddle equation gives $b_0(s) = s^{-\alpha}$, so $G_0(s) = M_0(\varepsilon s - 1/2)$. Moving from R_s to the maximizing integer changes $\mathcal{L}_0$ by $O_\alpha(\eta/R)$, since its first derivative vanishes and its second derivative is $O_\alpha(\eta/R)$ there. The other explicit terms have changes of the same or smaller order. Finally $\eta^2\varepsilon = (\eta/R)\varepsilon^2$. Equation (38) follows.

Expansion of the implicit root gives

$$s = 1 + \frac{2\varepsilon}{\alpha(\alpha-1)} + \frac{2(3-\alpha)\varepsilon^2}{\alpha^2(\alpha-1)^2} + O_\alpha(\varepsilon^3).$$

Substituting into (37) yields

$$E_0 = \frac{\alpha}{\alpha-1}\varepsilon - \frac{\alpha+1}{\alpha-1}\varepsilon^2 + O_\alpha(\varepsilon^3), \quad \eta M_0(\varepsilon s - 1/2) = -\eta/2 + O_\alpha(\eta\varepsilon),$$

which proves (39). $\square$

Corollary 9.7 (Three second-order regimes). *For the exact Zipf distribution,*

$$\begin{aligned} 1 < \alpha < 2: \quad L(\eta) &= \frac{\alpha}{\alpha-1}\varepsilon - \frac{\alpha+1}{\alpha-1}\varepsilon^2 + o(\varepsilon^2), \\ \alpha = 2: \quad L(\eta) &= 2\sqrt{C\eta} - \left(3C + \frac{1}{2}\right)\eta + O(\eta^{3/2}), \quad C = \frac{6}{\pi^2}, \\ \alpha > 2: \quad L(\eta) &= \frac{\alpha}{\alpha-1}\varepsilon - \frac{\eta}{2} + o(\eta). \end{aligned}$$

Proof. Compare $\varepsilon^2 = C^{2/\alpha}\eta^{2-2/\alpha}$ with η in (39). They exchange order at $\alpha = 2$, where both second-order contributions must be retained. $\square$

The quantified formula (39) is convenient across all exponents. It should not be read as resolving every displayed term separately when that term lies below its stated remainder. The implicit envelope avoids this issue by retaining an analytic function of ε without truncating its powers.

9.5 The Bernoulli correction from integer degree selection

The envelope admits a uniform refinement that records the optimal degree's position in the integer lattice. Write $B_2(u) = u^2 - u + 1/6$ for the second Bernoulli polynomial.

Theorem 9.8 (A lattice refinement for exact Zipf weights). *With the notation of Theorem 9.6,*

$$\begin{aligned} L(\eta) &= E_0 + \eta M_0 \left(\varepsilon s - \frac{1}{2} \right) + \frac{\alpha\eta}{2R} B_2(\{Rs\}) \\ &\quad + O_\alpha \left(\frac{\eta}{R} \left(\varepsilon + \frac{1}{R} \right) \right). \end{aligned} \tag{43}$$

The estimate holds through every crossover of the maximizing integer.

Proof. Use (42) with $k = Rs + d$, where (41) bounds d . Direct differentiation gives

$$\mathcal{L}'_0(Rs) = 0, \quad \mathcal{L}''_0(Rs) = \frac{\eta}{R} M_0 \left(\frac{\alpha}{s} + 2\varepsilon \right), \quad G'_0(s) = M_0 \left(\frac{\alpha}{2s} + 2\varepsilon \right).$$

Taylor expansion therefore makes the contribution of order η/R equal to

$$\frac{\eta}{R} M_0 \left[\left(\frac{\alpha}{2s} + \varepsilon \right) d^2 + \left(\frac{\alpha}{2s} + 2\varepsilon \right) d + \frac{\alpha}{12s} \right].$$

Since $s = 1 + O_\alpha(\varepsilon)$ and $M_0 = 1 + O_\alpha(\varepsilon)$, this is

$$\frac{\alpha\eta}{R} \left(\frac{d^2 + d}{2} + \frac{1}{12} \right) + O_\alpha(\eta\varepsilon/R).$$

The remaining Taylor terms are $O_\alpha(\eta/R^2)$, and the $\eta^2\varepsilon$ remainder in (42) is absorbed because it equals $(\eta/R)\varepsilon^2$.

Let $u = \{Rs\}$. Away from an $O_\alpha(\varepsilon)$ neighborhood of an integer, (41) gives $k = \lfloor Rs \rfloor$, hence $d = -u$ and

$$\frac{d^2 + d}{2} + \frac{1}{12} = \frac{1}{2} B_2(u).$$

Near a crossover, the selected integer may be the adjacent one. The endpoint identity $B_2(0) = B_2(1)$ bounds the resulting discrepancy by $O_\alpha(\varepsilon)$, already included in the error. The same reasoning covers either choice at an exact tie and proves (43). $\square$

This refinement separates three effects within one rigorously controlled expression: the analytic tail optimization, the first correction from Bernoulli noise and the discrete tail, and the oscillation caused by integer degree selection. The final oscillation has scale $\eta/R = C^{-1/\alpha} \eta^{1+1/\alpha}$, whereas its phase follows the implicitly determined optimum $Rs_\alpha(\varepsilon)$.

10 A general theorem for finitely supported perturbations

The compactness mechanism in Theorem 8.5 depends on finite perturbations occurring almost surely. Independence of individual noise bits is unnecessary.

Theorem 10.1 (Compact optimization for arbitrary finite perturbations). *Let D be any random vector in the countable group $\bigoplus_{\mathbb{N}} \mathbb{F}_2$, independent of $X \sim \mu$. For every $S \subseteq \mathbb{N}$, define*

$$q(S) = \mathbb{P}(|S \cap \text{supp } D| \text{ is odd}).$$

Then q is continuous on the compact space $2^{\mathbb{N}}$, and

$$\sup_{f \text{ } \mu\text{-measurable}} \mathbb{P}(f(X) \neq f(X \oplus D)) = \max_{S \subseteq \mathbb{N}} q(S).$$

The measurable supremum is attained exactly when a finite subset maximizes q . In ZFC the same value is a maximum over arbitrary total labels whose comparison event is measurable.

Proof. Let $A_n = \{\text{supp } D \subseteq [n]\}$. Then $\mathbb{P}(A_n) \rightarrow 1$, and

$$|q(S) - q(S \cap [n])| \leq \mathbb{P}(A_n^c)$$

uniformly in S . Thus q is a uniform limit of continuous cylinder functions. Compactness and density of finite subsets show $\max_S q(S) = \sup_{S \text{ finite}} q(S)$. The measurable claims follow from Theorem 3.1, including the strict-average argument when no finite maximizing subset exists.

For an arbitrary total label with measurable comparison event, take n sufficiently large that $\alpha_n = \mathbb{P}(A_n) > 0$, and condition on A_n . Every positive-probability D -section is measurable. Average these sections over the finite group of translations supported on $[n]$. Invariance preserves the integral, and each pointwise orbit average is the score of a finite Boolean function. Its value is bounded by $\max_{S \subseteq [n]} \mathbb{P}(|S \cap \text{supp } D| \text{ odd} \mid A_n)$. Then

$$\alpha_n \max_{S \subseteq [n]} \mathbb{P}(|S \cap \text{supp } D| \text{ odd} \mid A_n) \leq \max_{S \subseteq \mathbb{N}} q(S).$$

The unconditional score is therefore at most $\max_S q(S) + 1 - \alpha_n$; pass to the limit.

Finally, for any fixed subset S , the representative construction (18) changes sign under a finite translation D exactly when $|S \cap \text{supp } D|$ is odd. It has a measurable comparison event of probability $q(S)$. Choose a maximizing subset. $\square$

Corollary 10.2 (One parity function supplies all compact optimizers). *Suppose one total parity function $h : \mathcal{X} \rightarrow \{-1, 1\}$ exists. It suffices for the arbitrary-label attainment assertion in Theorem 10.1, simultaneously for every law of D .*

Proof. For each $S \subseteq \mathbb{N}$ define

$$h_S(x) = h(x \cdot \mathbf{1}_S),$$

where coordinates outside S are set to zero. A finite flip u changes h_S by $(-1)^{|S \cap \text{supp } u|}$, so its comparison score is $q(S)$. The nonempty compact maximizing set of q has a lexicographically least member: at each coordinate choose zero if that prefix still has a maximizing extension, and otherwise choose one. Nested compactness gives a maximizing limit. Apply h_S to this canonically selected subset. The same h works for all laws; no additional choice of labels is needed. $\square$

This theorem provides a reusable finite–infinite principle. Measurable labels use finite Walsh characters. Arbitrary componentwise labels can realize the characters indexed by all subsets of the countable coordinate set. Passing to the compact space of all subsets can add maximizers without increasing the supremum. No new independence or choice equivalence is inferred from this observation.

11 Exact computations and reproducibility

The archive includes two complementary checks. Neither is used to replace a proof of an infinite statement.

11.1 Exhausting all small Boolean rules

The program code/verify_parity.py uses only integer and rational arithmetic. It enumerates every Boolean truth table in dimensions 1, 2, 3, 4, independently constructs the channel’s transition probabilities, and computes the Walsh transform by integer butterflies. For each rule it compares direct transition scores with the spectral formula.

It also computes the minimum exact average decision-tree cost by the dynamic program

$$C_n(f) = \begin{cases} 0, & f \text{ constant,} \\ 1 + \min_{i \in [n]} \frac{C_{n-1}(f|_{x_i=0}) + C_{n-1}(f|_{x_i=1})}{2}, & f \text{ nonconstant.} \end{cases}$$

All choices of the first query are examined, so this is the exact optimum under uniform input, not a heuristic cost. The routine stores $2^n C_n(f)$ as an integer.

Dimension	Truth tables	Parameter cases	Rule-parameter pairs
1	4	8	32
2	16	8	128
3	256	8	2,048
4	65,536	8	524,288
Total	65,812		526,496

Every listed case passed the following comparisons: direct versus spectral score, finite optimum versus $\max_k w_k$, direct versus spectral total influence, influence versus minimum query cost, and score versus $G(C_n(f))$. The checks include boundary noise rates, uniform, geometric, and concentrated challenge distributions. They also verify the interpolation construction and exact crossover examples. The recorded data are in `data/verification.json`.

Example	First optimal degree	Exact optimal score
$n = 4, p_i = 1/4, \eta = 1/4$	3	17/32
$n = 4, p = (8, 4, 2, 1)/15, \eta = 1/4$	2	23/40
$p_i = 2^{-i}, \eta = 1/10$	3	173/250
$p_i = 2^{-i}, \eta = 1/100$	6	929079903231/10 ¹²

For the first row, degree four ties with degree three. This is why the table specifies the first optimal degree. The nonlinear example 6.2 is checked separately on its complete five-coordinate truth table by the `figure/example` script.

11.2 Numerical checks of asymptotic coefficients

The program `code/check_asymptotics.py` uses 65-digit Decimal arithmetic, exact rational Bernoulli numbers, and Euler–Maclaurin tails. It evaluates $\alpha = 1.3, 1.5, 2, 2.5, 3, 5$ at $\eta = 10^{-5}, 10^{-8}, 10^{-11}$. The recorded output is `data/asymptotics_check.txt`.

These evaluations check the signs, sizes, and lattice phase in Section 9. They are not interval certificates: the error bounds are proved analytically in the manuscript. In particular, the negative half term in the Zipf tail and the coefficient $-(3C + 1/2)\eta$ at $\alpha = 2$ were independently checked.

The figure source code `create_figures.py` regenerates the three standalone vector figures and a small example table. Figure generation uses NumPy and Matplotlib; the two mathematical check programs require only the Python standard library.

[write] The shipped layout and the intake reruns, 5 October 2026. In ProveIt the programs and their outputs keep their delivered names and bytes, in `code/`, `data/` and `figures/` of the report directory; the source is `article.tex` and `build.sh` is in `code/`. Three delivered behaviours matter for reruns, which should therefore be made on a copy (the report’s README gives the routes): `create_figures.py` writes into `figures/` and `data/examples.json` two levels above itself, so run in place it would overwrite shipped files; `verify_parity.py` writes `verification.json` in the working directory unless given `--output`; and `build.sh` changes to its own directory and runs `latexmk` on `noisy_parity.tex`, so it fails from `code/`. On copies (Python 3.14.4; NumPy and Matplotlib for the figure script) the intake found: `verify_parity.py` passes in every dimension (65,812 truth tables, 526,496 pairs, about 3 s) and its JSON equals `data/verification.json` except the fields `python_version` (3.12.14 in the delivery) and `elapsed_seconds`; `check_asymptotics.py` prints `data/asymptotics_check.txt` exactly; `create_figures.py` reports “Exact examples and nonlinear optimizer: PASS”, the enclosure 0.844268768560169857728257178... to 0.844268768560169857728257273... of the value in Remark 8.7,

and a JSON file equal to `data/examples.json`; its figure PDFs differ in bytes from the shipped ones (another Matplotlib version), which the article uses.

12 A proposed formalization path in ProveIt

The following are proposed proof modules, not files already present in the repository. Their dependency order separates the finite algebra from measure theory and from optional set-theoretic assumptions.

Layer	Mathematical obligations	Result exposed
Finite characters	Define the cube as $[n] \rightarrow \mathbb{F}_2$; prove orthogonality, finite inversion, Parseval, and the one-flip/product-noise eigenvalue formula.	A finite exact upper bound with a parity witness.
Finite optimization	Prove ordered-prefix maximality, the formula for d_k , the cutoff criterion, and concavity up to the maximum.	Certified degree selection and a rational finite checker.
Decision trees	Define exact evaluation, restrictions, revelation, expected cost, and pivotality. Prove influence is paid for by a query.	The finite inspection frontier and equality characterization.
Countable probability	Construct the fair product measure, use cylinder density in L^2 , and justify the absolutely convergent spectral sum.	Countable optimization, stability, and weak escape.
Asymptotic analysis	Prove tail estimates, threshold inversion, and uniform Taylor bounds with explicit remainder dependencies.	Geometric and Zipf laws, including the lattice term.
Choice and events	Treat total parity as an explicit hypothesis; formalize finite-orbit averaging and success-event measurability separately from label measurability.	The summable-noise classification with visible assumptions.

The inspected ProveIt declarations `prod_one_add_mul_pow`, `prod_one_sub_pow_eq_sum_thueMorseSign`, and `sum_pow_binaryWeight_eq_one_add_pow` in `ThueMorseBooleanCube.lean` are natural finite algebraic starting points [9]. They establish a useful binary-cube dictionary; they do not already establish the optimization or countable-measure results here. No Lean build was performed for this manuscript.

[write] Checked against the current tree, 5 October 2026. The three declarations exist, in namespace `Fabius`, with the statements quoted in Section 1.4, and the description above is accurate. For the first layer of the table, the repository also has, in `ThueMorseWalsh.lean` [12], the vanishing of every nontrivial Walsh character sum on the dyadic block and a masked character sum; with `sum_pow_binaryWeight_eq_one_add_pow` these give the product-noise factor $a^{|S|}$ after a substitution (Section 1.4). Finite Walsh inversion and Parseval for arbitrary functions, the one-flip eigenvalue, and every later layer are not in the repository. Nothing of this report is formalized.

A sensible first deliverable would be the finite theorem with rational p_i, η , an explicit optimal degree, and an exact evaluator for the witness. This would turn a finite computation into a proved certificate interface. The more difficult countable and choice layers should

retain separate interfaces so that classical choice is not silently imported into a measurable statement.

13 Further research questions and topics

These are proposed directions for the designed problem, rather than a list of independently verified open problems in the literature. Several have direct finite versions and meaningful infinite analogues.

- Q1. Biased configurations and the value of adaptation.** Replace the fair-bit measure by a biased product measure and use coordinate Markov transitions that preserve those marginals. The orthogonal product basis then has non-Boolean basis functions, so the parity simulation no longer directly realizes its spectral weights as labels. Determine exact two- and three-coordinate examples, and identify conditions under which adaptive inspection strictly improves the optimum over fixed coordinate sets.
- Q2. Simultaneous tests of three or more configurations.** Theorem 3.1 preserves every two-input XOR response, but higher correlations depend on products of different Fourier coefficients. Classify which multi-input acceptance predicates still admit a resource-preserving parity simulation. A small finite counterexample would be as informative as a positive criterion.
- Q3. All Boolean functions in the optimizing eigenspace.** With strict weights the classification is complete. With ties, Example 6.2 shows that nonlinear optima occur. Determine the extremal eigenspaces that contain such functions, classify their minimum inspection costs, and study how a small perturbation of the tied weights selects among them.
- Q4. Different costs for different coordinates.** If reading coordinate i costs $c_i > 0$, spectral simulation still costs $\sum_{i \in S} c_i$ on parity S , and weighted influence is bounded by expected paid cost. The remaining finite problem becomes optimization over subset mass and cost. Find tractable distribution families, exact certificates, and approximation schemes; in the infinite case, determine when a cost bound ensures attainment.
- Q5. Algorithms for correlated finite perturbations.** Theorem 10.1 settles the variational value for every finite-support perturbation law. What efficient representations of that law permit efficient maximization over S ? In finite dimension the spectral formula can still require exponentially many characters. Identify structural conditions on correlated faults that make the maximizing subset recoverable without exhaustive search.
- Q6. Nonsummable heterogeneous noise.** The homogeneous model has finite measurable optimizers; summable noise has the compact classification of Theorem 8.5. For $\eta_i \downarrow 0$ with $\sum_i \eta_i = \infty$, determine a sharp criterion for measurable attainment. The relevant objective on finite sets is still $(2p(S) - 1) \prod_{i \in S} (1 - 2\eta_i)$, but its extension to all subsets need not be continuous. This is a concrete intermediate regime.
- Q7. Recovering challenge tails from optimal failure.** For regularly varying p_k , the failure exponent determines $(\alpha - 1)/\alpha$. Formulate converse Tauberian statements: under what minimal monotonicity or regularity hypotheses does a regularly varying $L(\eta)$ force a regularly varying challenge tail? Which irregular tails can have the same leading optimal loss?
- Q8. Second-order tail perturbations.** Assume $p_k = Ck^{-\alpha}(1 + Dk^{-\beta} + o(k^{-\beta}))$. Determine how this correction competes with the ε^2 term, the order- η half correction, and the

Bernoulli lattice term. A useful answer would list the transition surfaces in (α, β) and give uniform expansions on those surfaces.

- Q9. Joint truncation and small-noise limits.** For weights p_i/P_n on $[n]$, study the regime in which $n/(C/\eta)^{1/\alpha}$ tends to a finite positive constant. This should interpolate between full-panel parity and the infinite power-law optimizer. Determine the limiting degree and loss profile, including boundary layers near resource saturation.
- Q10. Effective values versus effective optimizers.** With computable challenge weights and a computable summability modulus for the noise, the uniform error bound (16) gives a route to certified approximations of the optimal value. Determine the computational complexity of deciding whether a finite maximizing set exists, and what additional separation promise permits extracting one.
- Q11. Almost-sure optimality and the exact parity principle.** Over ZF+DC, determine the strength of the assertion that every law of an almost surely finite perturbation has an arbitrary total label with a measurable comparison event attaining the compact optimum. Corollary 10.2 shows that one total parity function already suffices, uniformly across all such laws. Does the converse follow already from almost-sure perfection in a single noiseless experiment with all $p_i > 0$? The issue is whether a rule satisfying the parity identities on a conull invariant set can be corrected on the remaining classes without additional choice. Any formulation must respect the separations already proved by Serafin [4].
- Q12. Proof certificates and asymptotic verification.** Formalize the finite optimizer and decision-tree inequality first, then formalize the spectral simulation on countable products. A separate project can turn the exact Zipf expansion into explicit numerical remainder bounds. Such certificates would distinguish arithmetic verification from analytic proof and would fit ProVeIt’s existing finite-cube infrastructure.

The most immediate mathematical follow-ups are the nonsummable heterogeneous-noise attainment criterion, the tied-weight Boolean classification, and converse asymptotic reconstruction. The most immediate formalization target is the exact rational finite game.

13.1 Further questions and research

[write] Added 5 October 2026, batch 114. Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the source that is not proved in full is stated as an open question with its source, the argument offered and what is missing. The source’s Q1–Q12 above are its own proposals (“proposed directions for the designed problem, rather than a list of independently verified open problems”); none is answered or advanced by the write, except where noted. What the report proves toward each, and what is missing:

- **Q1:** nothing is proved for biased input measures; Proposition 8.3 concerns only the measurability of total parity labels under a biased measure. Missing: the examples and a criterion for strict gain from adaptivity.
- **Q2:** Theorem 3.1 covers two inputs only. Missing: any classification or counterexample.
- **Q3:** Theorem 6.1 confines all optima to the eigenspace spanned by χ_S , $S \in \mathcal{M}$, and classifies them under strict weights; Example 6.2 shows nonlinear optima under ties. Missing: the Boolean functions of the tied eigenspaces, their costs, and selection under perturbation.
- **Q4:** the source’s bound “weighted influence is bounded by expected paid cost” follows from the per-coordinate form of Lemma 3.2 (multiply by c_i and sum), and the spectral simulation of parity S costs $\sum_{i \in S} c_i$. Missing: the resulting optimization over subset mass and cost, and infinite attainment.

- Q5: the value is Theorem 10.1. Missing: algorithms.
- Q6: settled for homogeneous noise (Theorems 2.2, 6.1) and summable noise (Theorem 8.5). Missing: a criterion when $\eta_i \downarrow 0$ and $\sum_i \eta_i = \infty$.
- Q7: Theorem 9.4 is the forward direction. Missing: any converse.
- Q8: Theorems 9.6 and 9.8 treat exact Zipf weights ($D = 0$). Missing: the competition with a correction $Dk^{-\beta}$.
- Q9: Proposition 7.3 and Theorem 9.2 treat other regimes. Missing: the joint limit.
- Q10: (16) gives certified approximation of the value. Missing: the complexity results.
- Q11: Corollary 10.2 gives the forward direction (one total parity function suffices for every law). Missing: the converse and the exact strength over ZF+DC; Serafin’s Theorem 1 [4] (read by the write, Section 1.4) is the separation any answer must respect.
- Q12: nothing of this report is formalized; the Lean ingredients that exist are listed in Section 1.4.

The write adds three questions. No claim of the source was found to be wrong.

Q13. [write] Deterministic trees under an average cost constraint. (From the paragraph after the proof of Theorem 2.2, which declines to identify the two optimizations.) For a deterministic exact procedure with $\mathbb{E}Q \leq c$, $G(c)$ is an upper bound (Corollary 3.3 and Jensen, as in that proof), attained by $\chi_{\lfloor c \rfloor}$ at integers $c \leq K$; Remark 5.2 proves it is not attained at non-integer $c < K$ under strictly decreasing weights, and that on a finite cube the deterministic optimum is then strictly smaller. Missing: that optimum (is it $w_{\lfloor c \rfloor}$, or can a non-parity tree with average cost between $\lfloor c \rfloor$ and c do better?), the tied-weight case, and whether on a countable cube the supremum equals $G(c)$.

Q14. [write] Stability uniformly in small noise. (From the paragraph after Theorem 6.3.) The gap $\gamma(\eta)$ tends to zero on a countable cube (Remark 6.4), so (8) degenerates as $\eta \downarrow 0$. For regularly varying weights the paragraph after Proposition 9.5 gives a degree-concentration statement on the scale ηR . Missing: a quantitative stability statement on the natural scale of the loss, for geometric and for general weights.

Q15. [write] External inputs and priority. The classical inputs are quoted, not re-proved: the spectral-sample and influence facts (O’Donnell), the measurable chromatic example (Conley–Miller, located by the source at the end of their Section 1), the Hurwitz-zeta expansion (DLMF 25.11.43) behind (40). The write read Geschke–Lubarsky–Rahn and Serafin (Section 1.4), not the other three. Priority for the designed problem and its assembled results is not established; the source’s search was targeted, not exhaustive.

[write] Independent check of the write, 6 October 2026. An adversarial check made by the intake after the write (commit b7d0d5e96), with its own code, found every addition of the write correct; no item needed a correction. It followed Remark 5.2 step by step (Jensen’s inequality for the concave piecewise-affine G and its equality case, the reduction to prefix sets, the Booleanness contradiction) and confirmed it on all 16,430 deterministic decision trees on $n = 3$ in exact rationals: for $p = (1/2, 3/10, 1/5)$ and $\eta = 1/20$ the best deterministic score under $\mathbb{E}Q \leq c$ is strictly below $G(c)$ at each of the eight non-integer values of c tested in $(0, 3)$ and equal to it at $c = 1, 2, 3$, and likewise at $\eta = 1/8$ for the values tested there. It re-derived Remark 6.4 (the witness $S = [K - 1] \cup \{j\}$ with gap in $(0, 2p_K)$, and $K_\eta \rightarrow \infty$) and checked it numerically for $p_i = 2^{-i}$ and $p_i = 6/(\pi^2 i^2)$. It confirmed the optimum $\max_k [1 + (1 - 2\eta)^k (2P_k - 1)]/2$ and the maximizers of Theorem 6.1 by brute force over all Boolean functions on $n \leq 4$ (65,536 at $n = 4$, computed from the transition matrix without the Fourier step), including a tie between $\pm\chi_{[2]}$ and $\pm\chi_{[3]}$. It confirmed the four declarations of `TrueMorseBooleanCube.lean` and the Walsh sums of `TrueMorseWalsh.lean` as described in Section 1.4, together with the substitution $u = -\eta/(1 - \eta)$ that turns the first into $\mathbb{E}\chi_S(B) = (1 - 2\eta)^{|S|}$; re-ran the searches behind the statement that nothing else in the repository bears on this report; and confirmed the two second-route pointers to Lemmas 49.1 and 64.1 of `measurable-box-games`.

A Assumption and claim ledger

Claim	Required hypotheses	Verification status
Spectral simulation, Theorem 3.1	Fair product input; same measurable Boolean rule twice; independent XOR perturbation.	Proved; classical Fourier machinery credited.
Exact cost frontier, Theorem 2.2	Homogeneous $0 < \eta < 1/2$; ordered positive challenge weights; shared seed; exact query algorithms.	Proved; exhaustive small finite checks.
Finite dependence of all optimizers, Theorem 6.1	Strictly positive homogeneous noise below $1/2$. Strict weights needed only for the signed-parity classification.	Proved; nonlinear tie example included.
Zero-noise nonattainment, Theorem 7.1	Infinitely many positive challenge probabilities; measurable labels.	Proved by a countable spectral average.
Parity and weak choice, Proposition 8.1	The restricted parity principle, or full AC as a sufficient assumption.	Classical, credited; no equivalence to full AC claimed.
Summable-noise classification, Theorem 8.5	$\sum_i \eta_i < \infty$, every $\eta_i < 1/2$; completed measures. Choice for the arbitrary-label construction.	Proved, including the upper bound for measurable events.
Geometric and power-law asymptotics	Fixed tail parameters; $\eta \downarrow 0$; exact Zipf weights where second terms are stated.	Proved; Decimal checks are supporting evidence only.
Historical priority	A wider literature review and external expert scrutiny.	Not established.
Lean/Rocq formalization	A separate implementation and kernel check.	Proposed; not supplied or claimed.

B Package and source audit

The main source is `noisy_parity.tex`. It contains the complete article, proofs, and bibliography without external section inputs. The three figure PDFs are in `figures/`. The final PDF is `noisy_parity.pdf`. The archive also contains the two verification programs, the figure/example program, their recorded results, `README.txt`, and `build.sh`. Run the latter from the package directory, or compile directly with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error noisy_parity.tex
```

The recommended check commands are

```
python3 code/verify_parity.py --output data/verification_rerun.json
python3 code/check_asymptotics.py
python3 code/create_figures.py
```

The first records a runtime and Python version, so those fields may differ between runs. The mathematical data use exact fractions. The asymptotic check has a fixed parameter grid and prints high-precision diagnostics.

Repository references are pinned to commit `b71545625fedc631edc2ead542abbbfa72eb6063`. The inspected nearby report was read for overlap and status, not imported as a proof

assumption. The external mathematical sources used are the author’s Boolean-analysis textbook, the cited parity and descriptive-set papers, and DLMF’s Euler–Maclaurin formulas. The requested LinkedIn URL did not yield usable full content; the affiliation-independent statement of Glazer’s interests comes from his public Epoch AI biography.

The research process independently checked the main finite formulas, the decision-tree cost inequality, the summable-noise event argument, and the delicate Zipf coefficients. These checks support the report’s internal consistency; they are not peer review or machine formalization. No repository modification or communication to any named mathematician was made for this deliverable.

[write] In ProveIt, 5 October 2026. This appendix describes the delivered package. In the repository the main source is `article.tex`, `noisy_parity.pdf` and `README.txt` are not shipped (the report’s own `article.pdf` and `README.md` replace them), and `build.sh` lies in `code/`, where it fails; the commands above work in the delivered layout only. Use the routes of the report’s README, on copies; see also the note at the end of Section 11.

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- [10] Vladimir Reshetnikov, ProveIt research-report collection. *Countable Information, Uncountably Many Boxes*, README, snapshot b71545625, inspected 5 October 2026. <https://github.com/VladimirReshetnikov/ProveIt/blob/b71545625fedc631edc2ead542abbfba72eb6063/SetTheory/Cardinals/docs/reports/ordinals-and-order-types/measurable-box-games/README.md>.
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- [12] [write] ProveIt repository. *The Walsh spectrum of the Thue–Morse block*, Analysis/FabiusFunction/Lean/FabiusFunction/ThueMorseWalsh.lean, read 5 October 2026 (unchanged since snapshot b71545625).